



Phonetic and phonological sound changes in an agent-based model

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ABSTRACT

The focus of the study is an agent-based model (ABM) concerned with simulating phonological stability and change using real speech data from a population of speakers. At the core of the model was a flexible and agent-specific association between acoustic exemplars and phonological categories. This was achieved by means of two general-purpose unsupervised machine learning algorithms: the first grouped exemplars into acoustic clusters, the second identified sets of clusters which largely contain exemplars of the same word classes. The model was tested first on data from Standard Southern British English, where a shift of /u/ to the front of the vowel space was expected not to cause any phonological re-categorisation. The simulation indeed showed phonological stability despite the phonetic change. Using the same settings, the ABM was then applied to data from New Zealand English in which a merger of /eə, ɪə/ diphthongs has taken place in the last 50 years. Compatibly with that, the simulation showed a shift from /eə/ towards /ɪə/ along with the neutralisation of the diphthong contrast. The results are discussed with respect to the conditions necessary to model a merger, taking into account both findings on marginal phonological relationships and principles from exemplar theory.

1. Introduction

The present study is about the development of a cognitively-inspired computational model, the interactive-phonetic agent-based model of sound change, called IP-ABM henceforth (Harrington et al., 2018; Harrington and Schiel, 2017; Stevens et al., 2019; Stevens and Harrington, 2022). Compatibly with many other computational models of sound change (Blevins and Wedel, 2009; Kirby, 2014; Sóskuthy, 2015; Stanford and Kenny, 2013; Todd et al., 2019; Wedel, 2006, 2012), the IP-ABM has a particular focus on explaining the origin and its association to the spread of sound change, given plausible hypotheses about the cognitive architecture that connects the speech signal in production and perception with some form of abstract phonological code and the lexicon. The IP-ABM of sound change is founded upon the idea that language usage carries the seeds of its change (Beckner et al., 2009). For this reason, agent interaction in which agents communicate with each other and update their linguistic knowledge is central to understanding how synchronic variation in language usage at the level of the individual is related to language change in the population.

The IP-ABM is informed by ideas developed over the last 20 years in episodic/exemplar (Johnson, 2006, 1997; Pierrehumbert, 2003, 2006) and usage models of speech (Bybee, 2001, 1999, 2010) in which there is a stochastic association between words, phonological units, and memorised speech signals that depends on experience, i.e. on the interlocutors to which the individual has been exposed over the

lifespan. As in other computational models inspired by episodic models of speech (Ettlinger, 2007; Sóskuthy, 2015; Todd et al., 2019; Wedel, 2006, 2004, 2007), one of the forces that drives sound change is that the sample of episodes that are activated in producing speech is a subset of the episodes that are updated by storing other talkers' episodes in speech perception. The evidence for this feedback connection between production and perception comes from a variety of sources including imitation and shadowing experiments (Goldinger, 1996, 2000) showing that under certain circumstances individuals imitate in production the phonetic attributes of talkers that they are exposed to (Babel, 2012; Babel et al., 2014), even in the absence of any social motivation to do so (Delvaux and Soquet, 2007; Nielsen, 2011, 2014; Pardo et al., 2012), and through longitudinal studies of speech showing phonetic changes in adults over their lifespan or as a consequence of living in another linguistic community for a long period of time (Harrington et al., 2000; Sancier and Fowler, 1997; de Leeuw, 2019; Riverin-Coutlée and Harrington, 2022; Sankoff, 2004).

The use of data in the IP-ABM departs from most other computational models of sound change in two ways. Firstly, the input to our model does not have to be artificially constructed, but can instead be real speech data recorded from actual speakers. The purpose of using real production data to initialise agents has been to ground the computational model in empirically observed phonetic variation, and also because this approach allows an exploration of how individual

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speaker characteristics, idiosyncrasies, and innovations are related to population change (Baker et al., 2011; Hall-Lew et al., 2021; Yu, 2021; François, 2014; Garrett and Johnson, 2011; Samuels, 1972; Yu and Zellou, 2019). Secondly, the production data used as input to the IP-ABM typically consists of parameters that capture characteristics of the time-varying speech signal instead of those that describe a static snapshot of speech (such as the temporal midpoint of vowels). The reason for this is because almost all of regular sound change originates from processes that are inherently dynamic such as coarticulation (Ohala, 1993; Lindblom et al., 1995; Beddor, 2009; Garrett and Johnson, 2011), undershoot (Beckman et al., 1992; Lindblom et al., 1995), or associations between events that are distributed in time that give rise to metathesis (Blevins and Garrett, 2004) and dissimilation (Alderete and Frisch, 2007; Abrego-Collier, 2013).

Agent-based models of sound change must be able to explain not only why change does occur but why it typically does not: that is the model's architecture should predict both change and stability (Sóskuthy, 2015; Stevens and Harrington, 2022; Wedel, 2007). Stability has been modelled as maintaining an equilibrium between the opposing forces of contrast maintenance (Blevins and Wedel, 2009; Blevins, 2005; Boersma and Hamann, 2008) and bias (Garrett and Johnson, 2013; Cohen Priva, 2012). Contrast maintenance is the pressure to keep words and the phonological code that differentiates them sufficiently distinct (de Boer, 2000) based on the idea that meaning in speech is communicated just as effectively after sound change has swept through the vocabulary as before (Ohala, 1989; Winter and Wedel, 2016). Recent evidence for contrast maintenance is that sound change tends to avoid creating homophones (Blevins and Wedel, 2009; Wedel, 2012; Wedel et al., 2013; Gessner and Hansson, 2004; Wedel and Fatkullin, 2017). Bias, on the other hand, is the tendency for speech sounds to change over time often because of physical limitations in producing and/or perceiving time-varying speech signals (Ohala, 1993). Bias tends to be magnified at points of low informativity (Cohen Priva, 2017), i.e. over stretches of the utterance that are less important for communicating the meaning of what is being said (Seyfarth, 2014). Sound change can then be modelled as shifting between two equilibria (Gafos and Kirov, 2009): the contrast between word meanings is typically maintained but there is a re-organisation in how words are mapped to speech signals via the phonological code. The finding that certain types of sound change might be influenced by how frequently words occur (Lin et al., 2014; Hay et al., 2015; Phillips, 2006) has recently been modelled in terms of the interplay between the opposing forces of contrast maintenance and bias (Todd et al., 2019).

Memorising exemplars of interlocutors is the main force in computational models of sound change that causes a category to expand and even broaden indefinitely. Holding the expansion in check is achieved in some computational models (de Boer, 2001; Kirby, 2014; Sóskuthy, 2015; Stanford and Kenny, 2013; Todd et al., 2019; Wedel, 2006, 2012; Kirby and Sonderegger, 2013; Wedel, 2007) and in the IP-ABM, by not memorising in speech perception exemplars that are either atypical for a category or not discriminable from another phonological category (Todd et al., 2019). Sampling exemplars that are most likely to be at the category's mean counters category broadening in speech production. The bias towards the mean in speech production has been explained in terms of entrenchment (Pierrehumbert, 2001; Bybee, 2007; Divjak and Caldwell-Harris, 2015) by which the same or similar speech production patterns are re-used. A further bias towards the mean would be additionally achieved if such episodes of a talker's own speech production were memorised (Winter and Wedel, 2016; Wedel, 2007). In the IP-ABM, this type of bias towards the mean comes about through sampling from a multidimensional Gaussian distribution over category exemplars. The category is also word-specific in the IP-ABM: that is, samples are drawn from separate distributions in producing /u/ from e.g. *food* and *soon*. This aspect of the model represents the idea that produced and perceived exemplars carry word-specific phonetic details (Pierrehumbert, 2002). In episodic models of speech, memorised

exemplars decay over time so they increasingly lose influence on the individual's production and perception of new tokens (Pierrehumbert, 2001). This idea has been implemented in computational models of sound change by means of assigning time-decaying activation levels to each exemplar (Wedel, 2006; Ettlinger, 2007; Kirby, 2013; Stanford and Kenny, 2013). The agents in the IP-ABM instead remove a random exemplar from their memory which is a less gradient implementation of time-decay.

In contrast to most other computational models of sound change, an agent's phonological knowledge is derived in the IP-ABM as a consequence of how word labels are stochastically associated with the memorised speech signals. This part of the IP-ABM is a simplified computational representation of the idea in Pierrehumbert (2003, p. 180) that “phonology does not abstract over speech directly, but rather indirectly via the abstraction of word forms”. Pierrehumbert (2003) also suggests that the phonological units are less abstract than taxonomic phonemes and are likely to be closer to some form of positional allophones. But since the number and nature of such categories are also unknown, the preference in the IP-ABM is to derive them algorithmically. The idea that phonology is not rigid but instead provides a flexible intermediate layer between word forms and memorised speech signals is central to deriving sound changes involving phonological splits and mergers in the IP-ABM (Stevens et al., 2019). As has been pointed out by Samuels (2006), a model of the kind in Labov (1994, p. 586) for explaining mergers can never result in a complete merger, if phonological categories repel each other as a consequence of rejecting ambiguous tokens.

Since with rare exceptions (e.g. Kirchner et al., 2010) exemplar models are vague about quite how phonological units emerge from the stochastic association between time-varying memorised speech signals and word labels, the IP-ABM draws upon some algorithms from the machine learning literature for this purpose which are nevertheless not without a cognitive foundation. As in the computational model of sound change by Kirby (2014), bottom-up processing makes use of Gaussian mixture models (Reynolds, 2009; Titterton et al., 1985) to derive acoustic classes from parameterised speech signals. GMMs have been shown to provide an effective simulation of the way that listeners combine and weight differently acoustic cues (Boersma et al., 2003; Francis and Nusbaum, 2002; Harmon et al., 2019; Idemaru and Holt, 2011; Toscano and McMurray, 2012) for the purposes of estimating category membership in speech perception in an unsupervised i.e. entirely bottom-up manner (Toscano and McMurray, 2010). Mixture models have also been used to explain how infants who lack prior knowledge of the number and nature of speech sound categories learn them bottom-up because of their ability to infer statistical distribution from experience (Erickson and Thiessen, 2015; Maye et al., 2008) i.e. from perceived speech signals (de Boer and Kuhl, 2003; Vallabha et al., 2007; see McMurray et al., 2009 for a further justification for the cognitive plausibility of modelling these types of categories as Gaussian mixture models). The top-down processing involves using a machine-learning technique known as non-negative matrix factorisation (NMF, Seung and Lee, 2001) in order to re-group the acoustic classes into phonological categories, depending on how word labels and acoustic classes are associated. This technique has been used to explain how semantic concepts are learned from multimodal input (Mangin et al., 2015) or how infants learn words from rich memories of perceived speech (Driesen et al., 2009; Mirzaei et al., 2013; Van hamme, 2011) without assuming any prior phonological knowledge.

Section 2 of this study briefly describes the architecture and the main processes implemented in the IP-ABM, while the interested reader is referred to Appendices A and B for a detailed exposition on the IP-ABM design and implementation. In the subsequent two sections, the model is tested to determine whether it can correctly predict the outcome of two types of sound changes in progress. One of these is /u/-fronting in Standard Southern British English that was previously analysed with an agent-based model in Harrington and Schiel (2017)

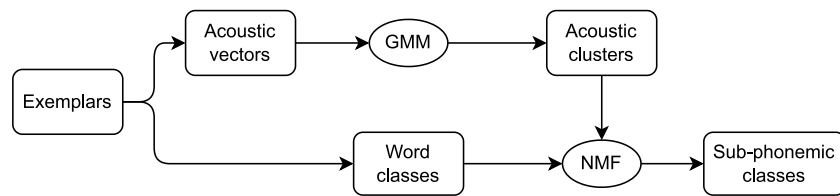


Fig. 1. General scheme of the derivation of acoustic and sub-phonemic representations.

but with fixed, pre-specified phonological categories. The purpose of re-applying the IP-ABM was therefore to test whether similar results are obtained when phonological classes are flexible and derived as a consequence of how word labels are associated with memorised speech signals. Section 4 then applies the IP-ABM to a sound change in progress in which phonological re-categorisation takes place in the *NEAR / SQUARE* centring diphthong merger of New Zealand English. Finally, a general discussion is reported in Section 5, where insight on the conditions necessary for a phonological merger to occur is provided with the help of artificial data.

2. The IP agent-based model

2.1. Agents and exemplars

In the IP-ABM each agent represents an individual speaker. All agents are identical, except for the speech material their memory is initialised with. There is no explicit model of the environment in which the agents interact, nor a notion of agent location. An agent is composed of three elements: a lexicon, a set of speech exemplars stored in memory, and a phonological model. The lexicon is fixed and identical for all agents. The exemplar memory determines an agent's state, which keeps changing during the simulation. The phonological model is periodically re-computed on the basis of the lexicon (fixed) and stored exemplars (variable).

Agents exchange and memorise isolated speech sounds represented by *tokens*. A token is defined as a pair (x, w) , where x is an acoustic representation, e.g. a set of formant values, and w is the associated lexical entry, which consists of the word type and, implicitly, the position within the word in which it was produced, e.g. the vowel in *food* (for an overview of symbols used for entities in the ABM, see Table 2). When a token is memorised by an agent it is called an *exemplar*.

2.2. Derivation of acoustic and sub-phonemic representations

Phonological classes are determined independently by each agent and are never communicated between agents. The computation of a phonological system takes place in two stages, depicted in Fig. 1. The first stage defines acoustic clusters, which are intermediate representations solely based on the acoustic information attached to exemplars. These are computed by applying GMM-based clustering to the entire set of stored exemplars in an agent's memory (see Appendix A.1 for a formal definition). For example, the left panel in Fig. 2 shows the formation of four clusters, each one represented by a GMM component.

In the second stage, acoustic clusters are combined together on the basis of their lexical content to form phonological classes. A phonological class consists of one or more acoustic clusters containing word classes that for the most part do not occur in any other acoustic cluster. This unsupervised, approximate mapping is computed by a clustering algorithm based on NMF (see Appendices A.2 and B.2). Fig. 2 illustrates a mapping from four acoustic clusters to three phonological classes. For example, since exemplars of w_2 and w_4 appear almost exclusively in clusters a_1 and a_2 , these are joined together to form phonological class p_1 . Formally, phonological classes are represented at two distinct levels: at the symbolic level, they induce a proper partition on the

lexicon, where each class is identified by the subset of word types it includes, e.g. $p_1 = \{w_2, w_4\}$; at the acoustic level, they are combinations of acoustic clusters, e.g. p_1 is a composition of a_1 and a_2 . Such a duality allows for inconsistencies between the two domains, such as for the two exemplars of w_2 and w_4 in Fig. 2 (blue) located inside the p_2 ellipse (yellow), i.e. lexically they belong to p_1 but they are acoustically closer to p_2 . A certain amount of acoustically ambiguous exemplars are tolerated (see *purity* in Appendix B.2) to reflect the stochastic nature of the relation between words, phonological classes, and exemplars. Each agent executes the two-stage computation of phonological classes first at initialisation and then periodically along a simulation, in order to allow its phonology to reflect the ever changing composition of memorised exemplars.

Understanding the relationship between the phonological classes created by the agents during the IP-ABM simulations and the classical phoneme (or s-phoneme using the terminology in Kiparsky, 2018) and its phonetic variants is important. There are three main cases to consider.

1. *Phonetic variants of different (classical) phonemes.* These are likely to fall into different phonemic classes in an agent's representation. This is because they map to mostly different sets of words (e.g. /i/ as in *feed*, *heed*, *weed* vs. /u/ as in *food*, *who'd*, *wooded*) and because they (in this case: /i/ vs. /u/) are acoustically distinct (and so likely to be in different acoustic clusters).
2. *Variants of the same classical phoneme that are in complementary distribution.* These are also likely to fall into different phonemic classes in the IP-ABM, even though they are not contrastive in the classical phonemic sense. Consider for example clear /l/ and dark /ɫ/ in many varieties of English. There are two reasons why these would in most cases be in different sub-phonemic classes. Firstly, they map to different sets of words (clear /l/ mostly to domain-initial position in e.g., *leaf*, *lot*, *luck*; dark /ɫ/ mostly to domain-final position in e.g., *call*, *full*, *sell*). Secondly, clear and dark /l/ have quite different acoustic characteristics (Kirkham, 2017; Lehiste, 1964; Sproat and Fujimura, 1993).
3. *Variants of the same classical phoneme that occur in the same context.* These include free variants (Reinisch et al., 2020): for example, the same speaker might produce word-final /t/ in *bat*, *let*, *sit* sometimes with a strong release/frication and sometimes with hardly any release and glottalisation (Penney et al., 2018; Seyfarth and Garellek, 2020). These variants are acoustically quite different and as a result will tend to fall in two different acoustic clusters. But because the clusters contain predominantly the same sets of words (with /t/ realised with frication vs. glottalisation respectively) they are likely to be grouped into the same phonemic class by an agent. This is the main case in which there is genuine abstraction in the IP-ABM, i.e. in which acoustically quite different variants are mapped to the same phonemic class.

There are many other types of phonetic variants that cannot be cleanly defined by the three cases detailed above, partly for the reason, as Scobbie and Stuart-Smith (2008, p. 87) note, that “contrast must be treated as a gradient phenomenon at the phonological level, with membership of a phonemic inventory being a matter of degree”.

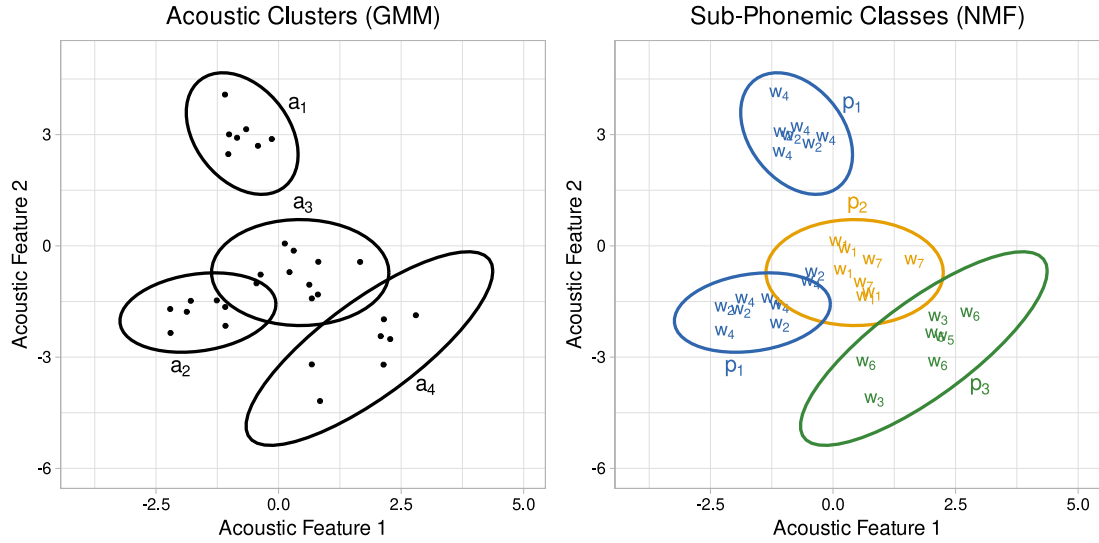


Fig. 2. Schematic representation of the two-staged derivation of phonology. Stage 1 (left) is the grouping of exemplars (black dots) on the basis of acoustic similarity, which produces acoustic clusters a_1 – a_4 (black ellipses). Stage 2 (right) is the grouping of acoustic clusters on the basis of lexical homogeneity (coloured ellipses). Here a_1 and a_2 , although acoustically different, are both populated by tokens of words w_2 and w_4 , which tend to not occur in the other clusters, so they are joined together forming sub-phonemic class $p_1 = \{w_2, w_4\}$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Thus whether or not some of the subtle quasi-phonemic distinctions documented by *Scobbie and Stuart-Smith (2008)* such as the diphthong differences in *days* vs. *daze* in Northern Irish English or the initial vowel differences in London English between $/a^w/$ in *molar* vs. $/ɒ^w/$ in *roller* fall into different phonemic classes depends entirely on the sensitivity of the acoustic parameterisation to such differences: if the acoustic differences are sufficiently large so that these variants are clustered separately, then they would also be in different phonemic classes (also because the phonemic classes map to different sets of words). On the basis of the above discussion, we will use the term *sub-phonemic classes* hereafter to refer to the algorithmically derived phonological classes in the IP-ABM.

An extended version of this section, including formal definitions, an expanded version of *Fig. 1* and a worked example based on a IP-ABM simulation is available in *Appendix B.2*.

2.3. Interaction between agents

A schematic outline of the interaction between an agent speaker and listener is shown in *Fig. 3*, where both panels represent entities in a common two-dimensional acoustic space in the respective agents' memories. The agent speaker randomly chooses a word class, w_1 , and builds a Gaussian model (black ellipse) over the acoustic representations of w_1 's exemplars stored in its memory. The agent speaker then samples an acoustic value x from that model to build a new token (x, w_1) (w_1 in red, where x encodes its position in the cartesian plane). The agent listener receives the token as is and looks up the phonological class with which w_1 is associated, in this case p_2 . The agent listener memorises the token if it passes both the discriminability and typicality test. The discriminability test is passed if the token's acoustic vector x is closer to the acoustic cluster(s) that make up the phonological class to which it belongs according to the lexicon (p_2) than to the acoustic clusters that make up the competitor classes (p_1 and p_3). The test is operationalised by computing the maximum a posteriori probability of x against a Mixture Discriminant Analysis model (MDA, *Fraley and Raftery, 2002*), a hierarchical model where each phonological class is a GMM (see *Appendices A.1* and *B.1*). The typicality test is passed if the token's acoustic vector x lies within a mean-centred boundary enclosing 95% of the probability mass of the distribution representing its phonological class (p_2 , see *Appendix B.1* for further details). If the exemplar is memorised, another exemplar chosen at random from the

same phonological class is deleted, provided that its deletion does not cause the loss of an entire word class. This memory loss is done to prevent the memory from enlarging linearly with every interaction and memorisation that takes place.

2.4. Initialisation and termination

Agents' memories are initialised with speech material from actual speakers: typically one speaker initialises one agent. Resampling techniques based on the SMOTE algorithm (*Chawla et al., 2002*) are applied whenever the speech material is insufficient to guarantee robust estimation of GMMs (see *Appendices A.3* and *B.3*). An agent population may be divided into groups representing different speaker communities or different stages of an attested sound change, the latter being the case for the present study, by initialising agent groups with speech material from corresponding speaker groups. Interactions between agents may be regulated accordingly by allowing only inter- or intra-group interactions.

Simulations are run until the agent population ceases to change and reaches an equilibrium. The nature of this equilibrium, as well as the path towards it, define the outcome of a simulation. Sound change manifests itself whenever an agent population of heterogeneous composition, initialised by speakers of different accents, evolves towards a common accent.¹ Stability is observed whenever agents retain their initial accent after prolonged interactions amongst each other. Both stability and change can be defined either at the acoustic-phonetic level, by observing the evolution in the acoustic properties of the exchanged speech tokens, or at the phonological level, by observing the evolution of the agent-specific phonological models across the population.

Like every model, the IP-ABM has limitations and contains implicit assumptions and simplifications, some of which are discussed in *Appendix B.4*.²

¹ We use the term accent to refer to a speaker's speaking style or idiolect which differs from another speaker's idiolect with respect to the sounds under investigation for reasons of e.g. regional origin or age.

² The IP-ABM was implemented as an R package called *soundChangerR* which is available at <https://github.com/IPSLMU/soundChangerR>. The version of the software used for the following simulations was 1.0.0.

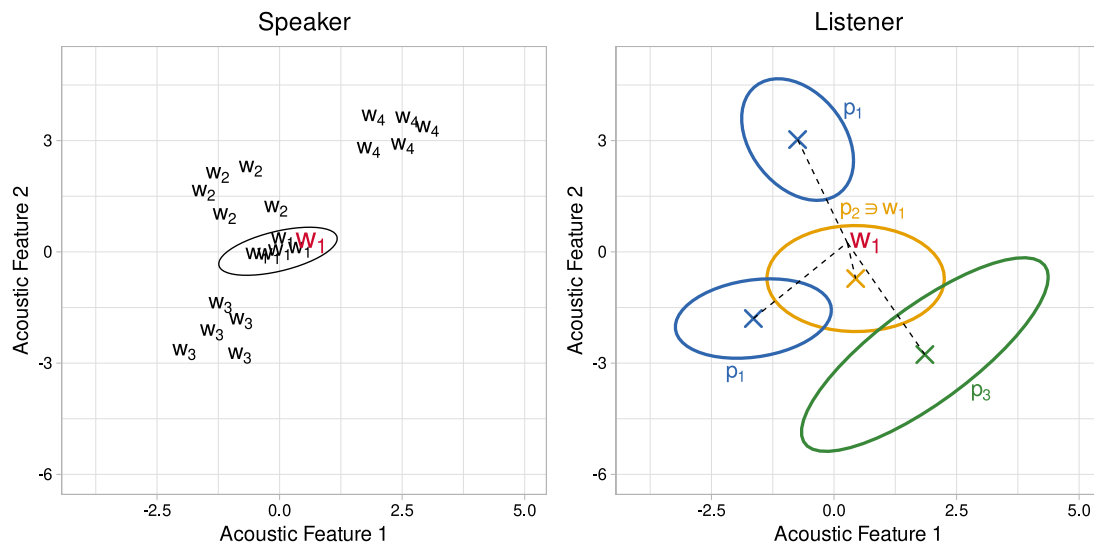


Fig. 3. Schematic representation of a token exchange between an agent speaker (left) and an agent listener (right). Ellipses represent mean-centred boundaries around Gaussian distributions enclosing 95% of their probability mass. The means are marked with a $\times$. The speaker produces a new token of a target sound of word type w_1 by estimating a Gaussian distribution based on all the stored exemplars of w_1 (black ellipse) and extracting a sample from it (in red). The listener receives the token as is and applies two memorisation tests based on its local phonological classes (coloured ellipses). The typicality test imposes a threshold on the maximum distance between the token and the phonological class it belongs to (here p_2); the discriminability test ensures that the token is closer to its phonological class than to the competing ones (here p_1 and p_3). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3. Simulating a phonetic shift

In this section, the IP-ABM is applied to /u/-fronting in Standard Southern British English (SSBE) that had previously been analysed by Harrington and Schiel (2017) with an earlier version of the model which did not allow for the derivation of sub-phonemic classes. The purpose of the study reported in this section is to investigate whether a similar result can be obtained with the modified architecture described in Section 2. It was important to check whether established results can be upheld with the IP-ABM and especially so if no phonological change was expected, as is the case for /u/-fronting in SSBE.

3.1. Introduction

The study reported by Harrington and Schiel (2017) was concerned with the effects of group contact on phonetic variants of the same phonological category. The general theory to be tested was that if there are two groups of speakers, A and B, that come into contact with each other, and if B's phonetic variants are skewed towards A's in some acoustic space, then when A and B communicate with each other, B's phonetic variants will shift towards A. The theory was tested using an agent-based model in Harrington and Schiel (2017) to simulate contact between two groups of SSBE speakers, one older and one younger, who produced GOOSE vowels with retracted and fronted variants, respectively. This difference between these two groups is a consequence of a sound change in progress in SSBE (Alderton, 2020; Bauer, 1985; Harrington et al., 2008; Hawkins and Midgley, 2005; Henton, 1983) which is also in evidence in many other varieties of English (Cox and Paley, 2001; Fridland, 2008; Gordon, 2004; Haddican et al., 2013; Jansen, 2018; Kerswill and Williams, 2005; Koops, 2010) by which variants of GOOSE have been shifting to the front of the vowel space in the last few decades. Older SSBE speakers in Harrington and Schiel (2017) had retracted variants of /u/ in neutral contexts such as *who* (in which there is a minimal coarticulatory influence from the preceding consonant) and in consonantal contexts that induce F2-lowering such as *boo* or *woo*. Older SSBE speakers also had fronted variants following either /j/ (*beauty*) or alveolars (*soon*, *too*). Younger speakers by contrast had fronted variants in all contexts, presumably because for them the /u/-fronting sound change was at an advanced stage (Harrington et al., 2008).

The prediction in Harrington and Schiel (2017) was that, because older speakers' /u/ with a back and a front variant was skewed towards younger speakers' /u/ with only a fronted variant, the shift should be asymmetric such that older speakers' /u/ shifts more towards younger speakers' /u/ than vice versa. This prediction was supported by the outcome of simulated interaction between members of the two groups using the agent-based model in Harrington and Schiel (2017, p. 430). Consistent with the original hypotheses by Harrington and Schiel (2017), the more specific predictions regarding the outcome of the interactions between agents representing younger and older SSBE speakers in the agent-based model were that:

1. Older agents /ju, u/ should shift more in the direction of younger agents' /ju, u/ than the other way round.
2. There should be no phonetic change to /i/ (given that it is not skewed in any direction).
3. There should be no phonological changes and no evidence of any splits or mergers.

3.2. Method

3.2.1. Data

The data used in the first experimental part of this paper are the same subset of the data from Harrington et al. (2008) that was used in Harrington and Schiel (2017). For convenience, we provide a brief description of the data here, but more details on the processing procedures can be found in Harrington and Schiel (2017). The data consists of utterances from 22 speakers of Standard Southern British English, equally divided into two age groups (younger: 19–21 years old, older: 56–88 years old). The speakers produced up to ten repetitions of eleven isolated monosyllabic words that contained either /i/ (e.g. *feed*) or /u/ (e.g. *food*) or /ju/ (e.g. *feud*; see Table 1 in Harrington and Schiel (2017) for the full list of words).³ The total number of productions was 11 (word classes) $\times$ 10 (repetitions) $\times$ 22 (speakers) = 2420. However, 30 of these were excluded due to mispronunciations, leaving 2390 productions. All /i, u, ju/ vowels were segmented between

³ The vowels /i, ju, u/ were all long vowels. We refrain from using the length mark for convenience.

their acoustic onset and offset without any further sub-segmentation of /ju/. Their first four formant frequencies were calculated using a frame shift of 5 ms and a 30 ms Blackman window. If necessary, formant tracking errors such as the misidentification of F3 for F2 in back vowels were manually corrected. The formant values in Hertz were converted to Bark using the formula in Traunmüller (1990). The formant trajectories were linearly time-normalised to 11 equally spaced data points between the acoustic vowel onset and offset.

3.2.2. Parameterisation

In contrast to Harrington and Schiel (2017) who used the Discrete Cosine Transform (DCT), the time-normalised F1 and F2 trajectories were parameterised using Functional Principal Component Analysis (FPCA, Ramsay and Silverman, 2005; Gubian et al., 2015) in this study. FPCA identifies the main modes of variation in functional, i.e. time-varying, data. The resulting curve parameterisation is usually more compact than the one obtained via DCT for two reasons. First, while DCT coefficients encode global characteristics of curves (i.e., mean, slope, curvature), PC scores are data-specific, i.e. their interpretation changes depending on the input data. Second, FPCA can handle multidimensional data. That is, F1 and F2 trajectories can be represented by e.g. just three Principal Component scores, each modulating both F1 and F2 instead of by six DCT coefficients (three each for F1 and F2, respectively).

The following formulae define the FPCA-based parameterisation of the formant trajectories:

$$F1_i(t) \approx \mu_{F1}(t) + \sum_{k=1}^K s_{k,i} \cdot PCk_{F1}(t) \quad (1a)$$

$$F2_i(t) \approx \mu_{F2}(t) + \sum_{k=1}^K s_{k,i} \cdot PCk_{F2}(t) \quad (1b)$$

$F1_i(t)$ and $F2_i(t)$ are the time-varying F1 and F2 trajectories respectively of the i th vowel production ($i = 1, 2, \dots, N = 2390$), $\mu_{F1}(t)$ and $\mu_{F2}(t)$ are the mean F1 and F2 trajectories, $PCk_{F1}(t)$ and $PCk_{F2}(t)$ are time-varying Principal Components (PCs, $k = 1, 2, \dots, K$), and $s_{k,i}$ are PC scores that modulate PCk and are different for each of the $N = 2390$ vowel productions. That is, the F1 and F2 trajectories are linearly decomposed into a set of PCs added to the mean trajectories. The PC scores are the same in (1a) and (1b) because each PC score modulates both F1 and F2 to capture possible systematic co-variations. The scores for the first $K = 3$ PCs were used as acoustic exemplars in the simulation, i.e. the agents' acoustic space was 3-dimensional.

3.2.3. Simulation setup

Simulations were based on a population of 22 agents, one for each of the 11 younger and 11 older participants described in Section 3.2.1. Each agent was initialised with the same lexicon of 11 word classes and with the acoustic exemplars of the corresponding speaker. The agents communicated across age groups (a younger with an older agent or vice versa) in a total of 500,000 interactions. This procedure was repeated 10 times (i.e. 10 runs) in order to test for the robustness of the results.

3.2.4. Quantification of change

The acoustic and sub-phonemic levels were analysed and interpreted separately. For each of them it was determined whether there had been any change in the *post-run* relative to the *baseline*. The baseline was the state of the agents after initialisation and before any interaction. The post-run was the state after the given number of interactions had taken place.

The analysis of the acoustic level was conducted in the domain of time-varying F1 and F2 trajectories. These were obtained by applying FPCA-based reconstruction formulae (1a) and (1b) to the memorised exemplars (consisting of the three PC scores s_1 , s_2 , and s_3). For quantifying acoustic changes, generalised additive mixed models (GAMMs) were applied separately to the reconstructed F1 and F2. The fixed

(smooth) factors were age group (younger, older), state (baseline, post-run), and canonical phoneme (/i, u, ju/). The random effects were agent (22 levels), word (11 levels), and run (10 levels). The full specification of GAMM equations, detailed hypotheses and statistical results can be found in section S1.1 in the Supplementary Material.

The sub-phonemic level was analysed using a number of indirect measurements. First, sub-phonemic similarity matrices S were computed. Given the agent memories at a certain point in time and given the set of word classes $\{w_i\}, i = 1, \dots, N_w$, S is a $N_w \times N_w$ symmetric matrix where $S[i, j]$ is the proportion of agents who grouped words w_i and w_j together in the same sub-phoneme (this shall be shown in more detail in Fig. 5). In simple terms, S provides a picture of the population-wise subdivision of the vocabulary into sub-phonemic classes: the closer its values are to 0 or 1 the more the agent-specific sub-phonemic classes agree with one another; vice versa, the closer to 0.5 the more diverse the sub-phonemic classes are across the population. S also provided a way to monitor the degree of consistency across separate independent simulation runs, which generally starts very high and decreases over time. Consistency across separate runs was quantified by computing the correlations between pairs of S matrices from different runs at the same point in simulation time. These correlations were computed using the Mantel test (Mantel, 1967), which takes into account the statistical dependencies induced by the structure of a similarity matrix.

Changes to phonological categorisation were determined by calculating the degree to which sub-phonemic classes matched canonical phonemes. This was done by calculating *canonical agreement* A , which we define as:

$$A = \frac{1}{N_w} \cdot \sum_p N_{\text{majority}}(p), \quad (2)$$

where N_w is the total number of word classes and $N_{\text{majority}}(p)$ is the number of unique word classes in the sub-phoneme p that belong to the canonical phoneme, which counts the largest number of unique word classes in p .⁴ For instance, suppose an agent defines three sub-phonemes p_1 – p_3 , where p_1 contains all 4 /i/ words, p_2 all 3 /ju/ words and also the /u/ word *soup* and p_3 the remaining 3 /u/ words (11 word classes in total). Then the sum in Eq. (2) equals $4 + 3 + 3$, hence $A = \frac{10}{11} \approx 0.9$. Along with agreement, the number of sub-phonemic classes N_p was computed too. This is because in limit cases agreement alone may convey the wrong picture. For instance, a perfect agreement of 1 can be trivially obtained by forming one sub-phonemic class per word class. The question to be considered was whether agreement and the number of sub-phonemic classes differed between baseline and post-run, and whether patterns differed between the two speaker groups.

The changes in agreement were tested with the following binomial regression mixed model:

$$\begin{aligned} \sum_p N_{\text{majority}}(p)_i &\sim \text{Binomial}(\theta_i, N_w) \\ \text{logit}(\theta_i) &= \beta_0 + \beta_{\text{gr}} \cdot \text{gr}_i + \beta_{\text{st}} \cdot \text{st}_i + \beta_{\text{gr:st}} \cdot \text{gr}_i \cdot \text{st}_i \\ &\quad + u_{\text{ag}[i],1} + u_{\text{ag}[i],2} \cdot \text{st}_i, \end{aligned} \quad (3)$$

where i is the running index on agents $\times$ states, gr and st are binary variables encoding group (older = 0, younger = 1) and state (baseline = 0, post-run = 1), $\text{ag}[i]$ identifies the agent corresponding to index i , and u is a two-dimensional Gaussian variable with zero mean which models a random intercept for agent (22 levels) and a random slope by

⁴ Using the notation introduced in Appendix B, $N_{\text{majority}}(p) = \max_q (|\{w : p(w) = p, q(w) = q\}|)$, where q indicates canonical phoneme. The definition in (2) corresponds to purity of clustering with respect to known categories, e.g. see the purity function in R package *NMF* (Gaujoux and Seoighe, 2010). We used the term agreement to avoid confusion with purity as defined in Appendix B.2.

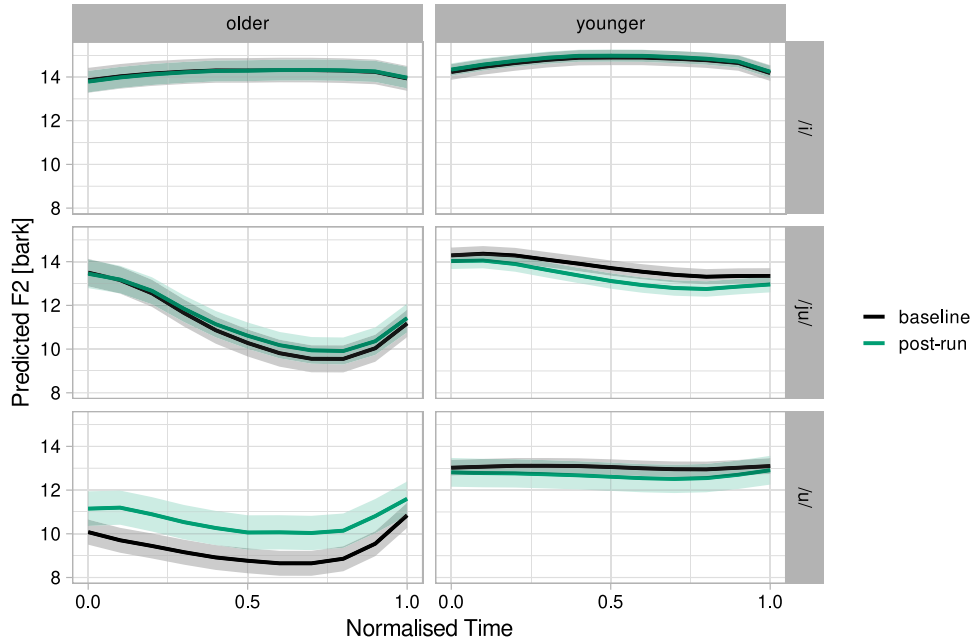


Fig. 4. Predicted F2 trajectories of /i/, /ju/, and /u/, both before (black) and after the simulation (green), with confidence bands. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

state (plus their correlation).⁵ Model (3) was applied on one randomly chosen simulation run out of the 10 computed, as opposed to using all of them and introducing run as further random factor, since the correlation between sub-phonemic similarity matrices S was always very high (>0.95).⁶

Similarly, changes in the number of sub-phonemic classes were tested. In this case the number of sub-phonemes minus one was modelled by a Poisson distribution⁷ whose parameter depends on group and state:

$$N_{p,i} - 1 \sim \text{Poisson}(\lambda_i)$$

$$\log(\lambda_i) = \beta_0 + \beta_{gx} \cdot gx_i + \beta_{st} \cdot st_i + \beta_{gx:st} \cdot gx_i \cdot st_i, \quad (4)$$

where i is the running index on agents $\times$ states.

3.3. Results

The estimated mean F2 trajectories in Fig. 4, which are more relevant to the sound change in question than F1, show a more fronted /u/ and a slightly more fronted /ju/ for older agents post-run (green) relative to the baseline (black). The same figure shows very little change for younger agents except for a slightly lower F2 after the simulation compared to before in both /u/ and /ju/. The results of the statistical analysis with GAMMs confirmed a significant shift of older agents' /u/ in the direction of younger agents' /u/ in F2 ($edf = 8.9$, $rdof = 8.9$, $F = 41.7$, $p < 0.001$).⁸ Contrary to the predictions, the

GAMMs showed a trend towards a significant retraction of younger agents' /u/ ($edf = 5.3$, $rdof = 5.4$, $F = 2.3$, $p = 0.05$). A test on the difference between these two changes confirmed that the fronting of older agents' /u/ was significantly larger than the retraction of younger agents' /u/ ($edf = 8.5$, $rdof = 8.6$, $F = 10.7$, $p < 0.001$). The shift from older agents' /ju/ towards that of younger agents was not significant, but younger agents' /ju/ shifted significantly towards that of older agents ($edf = 5.1$, $rdof = 5.2$, $F = 2.5$, $p < 0.05$). However, a test on the difference between these two changes showed no significance, i.e. there is insufficient evidence to show that the retraction of younger agents' /ju/ was greater than the fronting of older agents' /ju/. Compatibly with the predictions, the GAMMs showed no significant change to the F2 trajectory of /i/ for either age group.

The GAMMs computed for F1 (see section S1.1 in the Supplementary Material for visualisation) showed a significant raising of the trajectory for older agents' /u/ ($edf = 2.0$, $rdof = 2.0$, $F = 5.4$, $p < 0.01$), which is a change in the direction of younger agents' F1 trajectory for /u/, and no other significant change. In any case, the changes in F1 were always much smaller than those in F2.

Fig. 5 shows the association between the word and sub-phonemic classes, separately by age group (columns) and by state (before/after the simulation, rows), for one out of the ten runs. The darker a coloured tile, the higher the proportion of agents who grouped a given pair of words together in the same sub-phoneme. The three diagonal, dark purple squares in each of the four panels indicate that the association between exemplars and sub-phonemic classes was predictably related to canonical phonemes in both the baseline and post-run and for both age groups. That is, words containing /i/ were grouped almost exclusively with other /i/-words, and the same holds for /u/-words and /ju/-words. The main exception to this pattern was the word *soup* which was sometimes grouped with /ju/- instead of with /u/-words by older agents before the simulation, and with both /u/- and /ju/-words after the simulation. This undoubtedly came about because the high F2 locus of alveolars (Cooper et al., 1952; Iskarous et al., 2010) causes a strong acoustic resemblance to the high F2 onset and falling onset transition in /ju/-words.

The Poisson model in Eq. (4) and the corresponding post-hoc tests predicted 3.5 and 3.4 sub-phonemes for older, and 2.8 and 3.1 sub-phonemes for younger agents in the baseline and post-run, respectively.

⁵ Using R package *lme4* (Bates et al., 2015), Eq. (3) is specified as: `glmer(c(N_majority, N_words - N_majority) ~ gr * st + (st | ag), family = binomial, link = logit)`.

⁶ Using all runs and introducing run as further random factor in model (3) has only negative effects on convergence of estimation, precisely because the inter-run variance is too small.

⁷ The Poisson distribution models counts from zero to infinity, while the number of sub-phonemes N_p can vary between one and the number of word classes N_w . Subtracting one from N_p matches the minimum value to the Poisson distribution, while the upper limit is high enough ($N_w - 1 = 10$ in the SSBE case) to be ignored in the model.

⁸ edf stands for the estimated degrees of freedom, $rdof$ for the reference degrees of freedom.

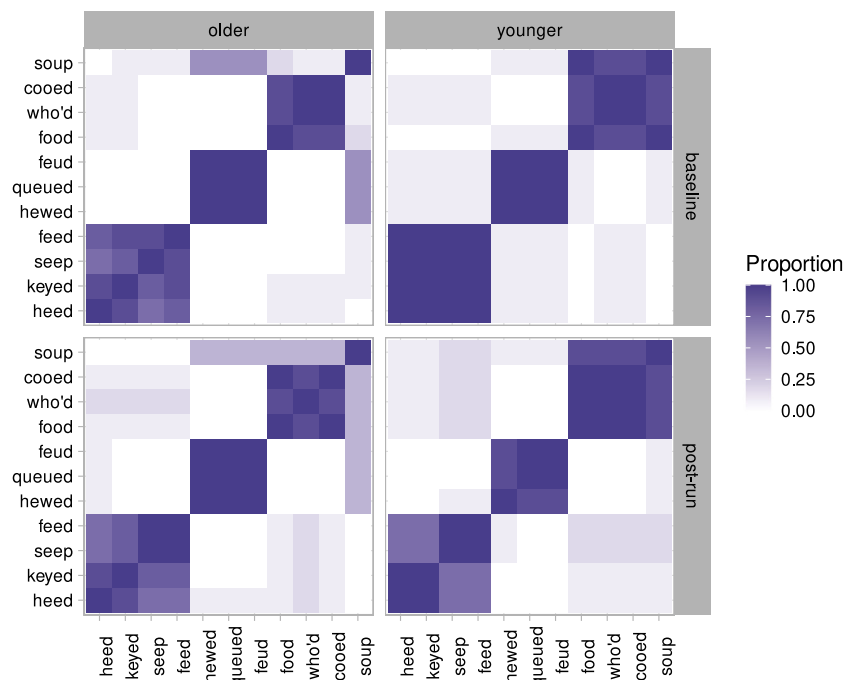


Fig. 5. Proportion of agents (colour-coded) who grouped together two word types in one sub-phoneme. The words on the axes are ordered by canonical phonemes (first /i/, then /ju/, and lastly /u/). This is the result for a single run of this simulation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

This shows that the number of sub-phonemes was close to the actual number of canonical phonemes in both age groups and both before and after the simulation, even though the agents were never given any information about the canonical phonemes. Neither age group nor simulation state had a significant effect on the number of sub-phonemes.

The expected canonical agreement (Eq. (2)) predicted by the generalised mixed model in Eq. (3) was 0.95 and 0.94 for older, and 0.98 and 0.96 for younger agents before and after the simulation, respectively. These results and subsequent post-hoc tests reveal the following three main findings. Firstly, and consistently with hypothesis 3, agreement was unaffected by state: that is, the degree to which the mapping between exemplars and sub-phonemic classes was related to canonical phonemes was no different in the baseline compared to post-run. Secondly, there were no significant differences between younger and older agents' canonical agreement. Thirdly, and perhaps most importantly, the predicted value of agreement was very high (≥ 0.94) in all cases. Thus overall, although the mapping from exemplars to sub-phonemic classes was completely unsupervised, it was nevertheless very strongly in agreement with the division of exemplars into canonical phonemes.

3.4. Discussion

The purpose of the analysis in this section was to compare the agent-based model described in Section 2 with that of Harrington and Schiel (2017) when applied to the same acoustic data. The application of the IP-ABM to this dataset differed from that of Harrington and Schiel (2017) in three major ways: phonological classes were derived in a data-driven way rather than being fixed; the acoustic parameters were based on time-varying F1 and F2 together rather than on just F2; and there was a different strategy for deleting exemplars from memory.

The results for the acoustic changes due to this agent-based model are broadly in agreement with those of Harrington and Schiel (2017). In particular, in both models there was an asymmetric shift such that older agents' /u/ shifted more towards that of younger agents than vice versa. Both models were also in agreement in showing no change to

/i/ in either older or younger agents. The main difference between the outcomes of agent interaction in the two models was in /ju/ which, in contrast to the model in 2, showed greater F2-raising for older agents and no significant F2-lowering for younger agents in Harrington and Schiel (2017). Visually, however, there was some F2-lowering in younger agents' /ju/, as can be seen from Fig. 8 in Harrington and Schiel (2017). Both in Harrington and Schiel (2017) and in the model tested here, this unexpected change in younger agents' /ju/ most likely comes about because of a repulsion effect between /i/ and /ju/ which are very similar acoustically. Hence, the relative memorisation criterion prevents acceptance of exemplars in the ambiguous acoustic space between /i/ and /ju/ and allows only the memorisation of /ju/-exemplars with lower F2.

In Harrington and Schiel (2017) the phonological classes were pre-assigned and immutable canonical (classical) phonemes (/i, ju, u/). The phonological classes in the present model were by contrast derived in an entirely data-driven way: above all, the derived sub-phonemic classes could split and merge during interaction (similarly to Stevens et al., 2019). Despite these very different phonological architectures, there was close agreement between the two models: in particular, in the IP-ABM the information contained in the derived sub-phonemic classes corresponded almost entirely with the division of the exemplar space into classical phonemes.

These two null results – the lack of acoustic change in /i/ and the consistency in phonological classification from initialisation to the end of the run – show that neither phonetic nor phonological change is an inevitable consequence of the agent-based model tested here. An agent-based model that always resulted in change would not be compatible with the evidence showing that it is only a fraction of phonetic variation that actually becomes sound change (Sóskuthy, 2015; Baker et al., 2011; Ohala, 1993).

4. Simulating a merger

This study was concerned with testing whether the same ABM architecture used to simulate the phonetic shift of SSBE speakers towards

a fronted GOOSE vowel in Section 3 could be employed to predict the merger of the lexical sets NEAR and SQUARE in New Zealand English. In contrast to /u/-fronting in SSBE, the diphthong merger in NZE has both acoustic and phonological ramifications both of which should be predicted by the IP-ABM given that it provides the possibility of re-organising the agents' phonological level.

4.1. Introduction

The merger of /ɪə/ and /eə/ in NZE was first described roughly half a century ago, based largely on auditory impressions (Hawkins, 1973; Turner, 1966; Wells, 1982; Turner, 1970). Later studies found that the merger had begun long before the 1970s, possibly even in the 19th century (Gordon and Trudgill, 1999; Gordon and MacLagan, 1985). Today, the /ɪə, eə/ merger is considered a characteristic feature of NZE in contrast with RP (Gordon, 2004) and Australian English (Holmes, 1995; Watson et al., 1998), and has been the subject of many quantitative acoustic analyses in the last decades.

In their seminal research project called “The Canterbury NEAR / SQUARE study”, Gordon and MacLagan (1990, 1985, 2001), MacLagan and Gordon (1996) recorded groups of 14-year old NZE speakers in 1983, 1988, 1993, and 1998. Over this time span there was a significant increase in the teenagers' usage of a merged diphthong for both NEAR and SQUARE. The spread of the merger amongst the teenagers followed a sigmoidal shape: Only 16% of all participants in 1983 and 22% in 1988 produced the merger consistently, but in the five years between 1988 and 1993 the merger suddenly took hold in the speech of 76% of all recorded teenagers. The data from 1998 showed a consolidation of this trend, i.e. 80% of the participants produced a merged variant. Gordon and MacLagan (2001) were also able to observe a change in the direction of the merger in their longitudinal data: While 32% of teenagers produced a diphthong closer to /eə/ and 19% produced a diphthong closer to /ɪə/ in 1983,⁹ the reverse was the case in 1988, and by 1998, only 1% of participants produced /eə/ for both NEAR and SQUARE, while 71% produced /ɪə/.

The NZE centring diphthongs were characterised by a high degree of acoustic variability for a long time before the merger was documented for the whole population (e.g., consider the contradictory accounts by Turner, 1970; Watson et al., 2000; Trudgill and Hannah, 1994). There is now abundant evidence showing that the first element of /eə/ was gradually raised such that most contemporary NZE speakers realise both ‘hear’ and ‘hair’ as /hɪə/ (Hay et al., 2008; Gordon and MacLagan, 2000; Rae and Warren, 2002; MacLagan and Hay, 2007; Hay et al., 2006; Kennedy, 2004; Warren and Hay, 2006; Watson et al., 1998). An acoustic analysis of a larger set of the same data to be used in the present study showed that younger NZE speakers made hardly any distinction between /ɪə/ and /eə/ (Gubian et al., 2019). Older speakers, however, still produced a slightly more peripheral /eə/ (i.e., with lower F1 and higher F2), thereby maintaining some acoustic separation between the diphthongs. Given the apparent-time approach, the results from that study provided support for the direction of change from /eə/ towards /ɪə/. Gubian et al. (2019) also found that the merger was accompanied by an increase in formant tilt, i.e. a change from a more monophthongal /eə/ towards a more diphthongal /ɪə/ (also see MacLagan and Gordon, 1996).

The study presented in this section aims to model the diphthong merger by means of interactions between two agent groups. Similarly to the study in Section 3, younger NZE speakers represent a more advanced or even completed stage of the sound change whereas older speakers represent an older state of NZE in which a contrast between /ɪə/ and /eə/ still existed. It was predicted that the interactions between younger and older NZE speakers should result in an acoustic and phonological merger of /ɪə/ and /eə/. The more specific hypotheses to be tested were that:

1. Older agents' /eə/ should shift more in the direction of younger agents' /ɪə/ than the other way round.
2. There should be no phonetic change to /ɪə/.
3. The sub-phonemes should agree less with the canonical separation of /ɪə/ and /eə/ after than before the simulation, especially amongst older agents.

4.2. Method

4.2.1. Data & parameterisation

The material and speakers are a subset of the data used in Gubian et al. (2019) which was taken from the New Zealand Spoken English Database recorded in 1999/2000. There were 48 speakers, equally divided into two age groups (older: 46–60 years, younger: 18–30 years), two regional groups (Hamilton, Wellington) as well as two genders (male, female). These speakers had produced 53 words once or twice in isolation or short sentences. There were 30 /eə/ and 23 /ɪə/ word classes. With the exception of *questionnaire*, *fun-fair*, and *thoroughfare*, the diphthongs occurred in the syllable with primary lexical stress. There were 1245 productions of /ɪə/ and 1530 productions of /eə/, resulting in a total of 2775 diphthongs to be analysed.

The words were automatically segmented using MAUS (Kisler et al., 2017) and the first five formant frequencies were computed using a frame shift of 6.25 ms and a Gaussian window of 25 ms. The segment boundaries and formant frequencies were manually corrected if necessary. The F1 and F2 trajectories of /ɪə/ and /eə/ were z-score normalised with respect to the speaker's /i, ɔ, ʊ/ and linearly time-normalised to 11 time points. Subsequently, the formant trajectories were parameterised by means of FPCA, as described in Section 3.2.2. The interpretation of the first $K = 3$ Principal Components for this New Zealand English dataset is described in Gubian et al. (2019).

4.2.2. Simulation setup

The IP-ABM was applied to the NZE data using the same settings as were used to simulate the phonetic shift in Section 3. That is, there were 48 agents, 24 of them representing the younger and 24 representing the older speakers. After initialising each agent with the word classes and exemplars of one of the speakers, SMOTE was applied to increase the number of tokens per word (see Appendices A.3 and B.3 for more details). The application of SMOTE to each agent resulted in a tenfold increase in the number of tokens per word. The agents interacted with agents of the opposite age group in a total of 300,000 interactions. The simulation was repeated 10 times.

4.3. Results

The predicted F1 trajectories in Fig. 6 show that older agents had a more peripheral /eə/ post-run (green) compared to the baseline (black) as shown by the substantially lowered F1 trajectory. There were hardly any visual changes in younger agents' diphthongs and to older agents' /ɪə/ apart from a slight reduction in tilt. In line with hypothesis 1, the GAMMs confirmed a significant change in older agents' /eə/ ($edf = 5.0$, $rd f = 5.3$, $F = 4.0$, $p < 0.01$), while the change in younger agents' /eə/ was not significant ($edf = 2.0$, $rd f = 2.0$, $F = 2.8$, $p = 0.06$). However, a test of the difference between those two changes did not reach significance, although the predicted trend goes in the expected direction: that is, the change in older speakers was greater than in younger speakers. The GAMMs also showed that the age groups differed significantly in their pronunciation of /eə/ before the simulation ($edf = 5.8$, $rd f = 6.0$, $F = 6.4$, $p < 0.001$), but were not statistically different after the simulation for either diphthong. Contrary to the predictions, the change in older agents' /ɪə/ was also significant ($edf = 2.5$, $rd f = 2.6$, $F = 5.1$, $p < 0.01$).

The GAMMs computed for F2 (see section S1.2 in the Supplementary Material for visualisation) showed that there was a significant difference between the two age groups' /eə/ before ($edf = 4.6$, $rd f = 4.9$,

⁹ The remaining 49% of teenagers produced a distinction between NEAR and SQUARE.

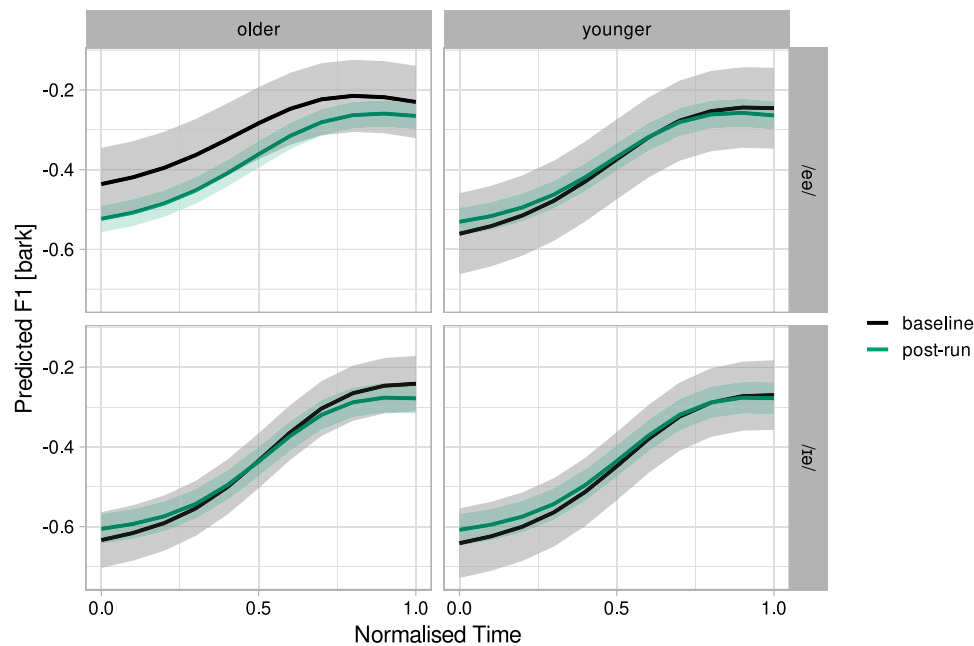


Fig. 6. Predicted F1 trajectories of /ɪə/ and /eə/, both before (black) and after the simulation (green), with confidence bands. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

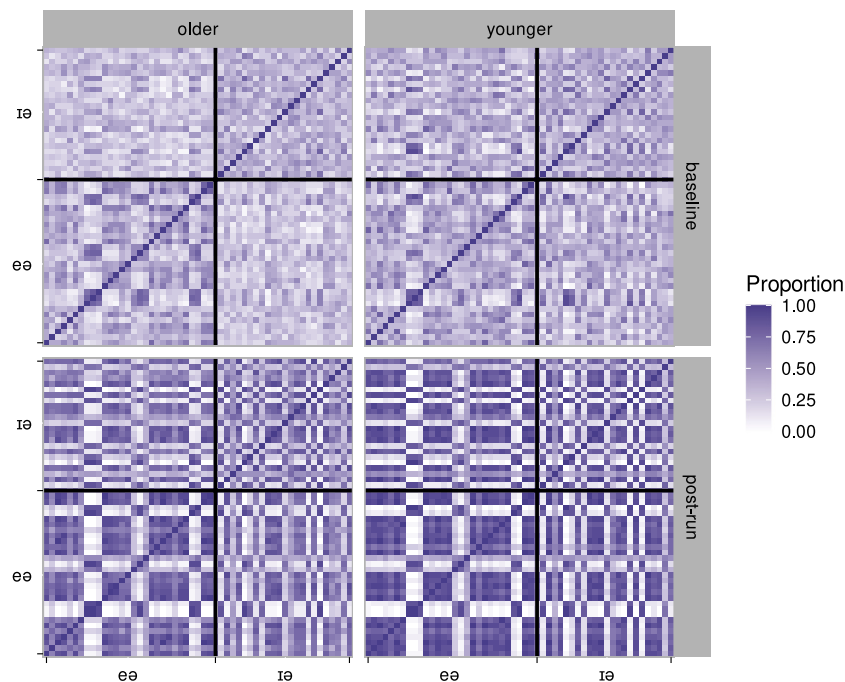


Fig. 7. Proportion of agents (colour-coded) who grouped together two word types in one sub-phoneme for one out of ten runs of this simulation. The 53 word types on the axes are ordered by canonical phonemes as indicated (the word labels are not shown for reasons of clarity). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

$F = 5.7$, $p < 0.001$), but not after the simulation. Furthermore there were some minor, yet significant changes in F2 tilt for younger agents' /ɪə/ ($edf = 2.1$, $rd f = 2.1$, $F = 4.6$, $p < 0.05$) and /eə/ ($edf = 2.0$, $rd f = 2.0$, $F = 3.9$, $p < 0.05$) as well as older agents' /ɪə/ ($edf = 2.0$, $rd f = 2.0$, $F = 5.9$, $p < 0.01$). The full specification of GAMM equations, detailed hypotheses and statistical results can be found in section S1.2 in the Supplementary Material.

For one out of the ten simulation runs, Fig. 7 shows the proportion of agents who grouped pairs of words into a sub-phoneme. The 53 words on either axis are not shown and are instead grouped

by canonical phonemes. If agents grouped words together in sub-phonemes according to the canonical distinction between /ɪə/ and /eə/, the panels would show two large dark purple squares, one on the bottom left and the other on the top right. Before the agents started communicating with each other (top row), younger agents did not show any discernible pattern of grouping words according to the canonical (classical) phonemes /ɪə/ and /eə/. Older agents, on the other hand, showed a mild trend towards grouping words according to classical phonemes, i.e. the squares on the bottom left and top right are slightly darker overall than the squares in the top left and bottom right. In the

Table 1

Word groups discernible from the bottom row of Fig. 7. Five out of the 53 target words were not grouped consistently enough by all agents to appear in this table. Where necessary, the word following the target word is shown as well. For each target word, the diphthong and following consonant were phonemically transcribed.

Group 1: before alveolars			
veers	/ɪəz/	shared	/eəd/
fierce	/ɪəs/	compared	/eəd/
scarcely	/eəs/	here to	/ɪət/
cares	/eəz/	bear to	/eət/
cheered	/ɪəd/	dare to	/eət/
beard	/ɪəd/	sincere condolences*	/ɪək/
appeared	/ɪəd/	prepare*	/eə/
chaired	/eəd/	merely*	/ɪəl/
Group 2: in final position			
dear old	/ɪəɪ/	experiences	/ɪəɪ/
sheer fluke	/ɪəf/	nearer	/ɪəɪ/
gear half	/ɪəh/	staring	/eəɪ/
Lear was	/ɪəw/	Hungarian	/eəɪ/
sphere	/ɪə/	Mary	/eəɪ/
fear	/ɪə/	hilarious	/eəɪ/
year at, year I	/ɪəɪ/	airing	/eəɪ/
jeer at	/ɪəɪ/	tearing	/eəɪ/
weir	/ɪə/	varies	/eəɪ/
fun-fair is	/eəɪ/	parents	/eəɪ/
questionnaire about	/eəɪ/	area	/eəɪ/
rare giant	/eəɪ/	various	/eəɪ/
wear black	/eəb/	fairly	/eəɪ/
aware that	/eəð/	heirloom	/eəɪ/
declare	/eə/	years*	/ɪəz/
mysterious	/ɪəɪ/	arrear*	/ɪəz/

Asterisks indicate words that are in the wrong category according to their transcription.

post-run there was a clear grouping pattern which was the same in both age groups and indicates that agents formed mainly two sub-phonemes which bore no relationship to the separation into canonical phonemes.

The generalised mixed model in Eq. (3) and subsequent post-hoc tests with *emmeans* showed that the predicted agreement between sub-phonemes and canonical phonemes was 0.68 for older agents and 0.61 for younger agents before the simulation, i.e. older agents adhered more to the canonical phonological separation than younger agents. In the post-run, the predicted agreement was 0.61 for older and 0.58 for younger agents, showing that the canonical agreement decreased in both age groups. The canonical agreement was overall much lower here than in the SSBE simulation and younger NZE agents reached almost chance level in the post-run.¹⁰ Importantly, this means that the agents incrementally generated sub-phonemes that contained a mixture of /eə/- and /ɪə/-words. In line with hypothesis 3, the contrast in canonical agreement between older and younger agents was significant before ($z = 3.0$, $p < 0.01$), but not after the simulation. There was also a significant contrast between the baseline and post-run for older ($z = 3.3$, $p < 0.001$), but not for younger agents.

The Poisson model in Eq. (4) and corresponding post-hoc tests predicted 3.3 and 2.3 sub-phonemes for older, and 2.8 and 2.2 sub-phonemes for younger agents before and after the simulation, respectively. That is, the number of sub-phonemes decreased in both age groups over the course of the interactions from roughly three to two sub-phonemes. The difference in number of sub-phonemes between the baseline and post-run was significant for older ($z = 2.7$, $p < 0.01$), but not for younger agents. In combination with the decrease in canonical agreement, these findings indicate a phonological merger of the two diphthongs under investigation.

¹⁰ Since 30 out of 53 word types in the dataset contained /eə/, the chance level, where a sub-phoneme contained a balanced mixture of /eə/- and /ɪə/-words, is $30/53 \approx 0.57$.

A closer investigation of the post-run word grouping into sub-phonemes (Fig. 7, bottom row) showed that the majority of agents formed two sub-phonemes which coherently consisted of exemplars from 48 out of 53 word classes. Table 1 shows that one sub-phoneme contained words in which the diphthong predominantly appeared before /d/, /t/, /z/, or /s/, and the other contained words in which the diphthong appeared in word- or syllable-final position. In order to quantify the emergence of this grouping pattern, formula (2) was adapted to compute the agreement between sub-phonemic classes and the two groups in Table 1 and the generalised linear mixed model from Eq. (3) was re-computed to predict the effect of age group and simulation state on the adapted agreement measure. As could be expected from Fig. 7, the model showed a significant effect of state ($z = 6.3$, $p < 0.001$), but not of age group on the adapted agreement. The predicted agreement between sub-phonemes and the context-dependent word groups was 0.78 for older and younger agents before the simulation, and 0.91 for older and 0.93 for younger agents in the post-run.¹¹ That is, the influence of the diphthongs' phonetic context (i.e., before alveolars or in final position) on how agents grouped word classes into sub-phonemes, while present from the beginning, increased significantly over the course of the simulation.

4.4. Discussion

The aim of the simulation presented in this section was to test whether the IP-ABM was capable of predicting the NZE NEAR / SQUARE merger using the same settings that were shown to successfully model /u/-fronting in SSBE. The results showed an asymmetric acoustic change in older agents' /eə/ towards that of younger agents. That is, the acoustic merger of the two diphthongs, which was close to completion for younger agents in the baseline, also took hold in the speech of older agents after they had conversed with younger ones. The statistical analysis of the acoustic change provided only a partial confirmation of the expected asymmetry. This may be because the predicted differences and changes were at most around 0.1 barks in size, which is an order of magnitude smaller than those predicted for the SSBE case. This means that differences between younger and older agents' acoustic exemplars were already quite small from the start, which makes it hard for clear patterns to emerge from noise (and be detected statistically). Nevertheless, the older agents' acoustic change from /eə/ towards /ɪə/ provides support for the hypothesis put forward by others (Rae and Warren, 2002; Hay et al., 2006; Gordon and MacLagan, 2001) that the NZE NEAR / SQUARE merger is a merger by approximation.

Furthermore, the regular and unsupervised updates of the agent-specific association between exemplars and sub-phonemes resulted in an increased neutralisation of the phonological contrast between the diphthongs over simulation time. This was most apparent again in older agents who, in the beginning, had sub-phonemes that somewhat mirrored the canonical separation of /ɪə/ and /eə/. In addition to the relatively small acoustic changes summarised above, the low canonical agreement for NZE as compared to SSBE agents at the baseline indicates that the merger was quite advanced for most of the NZE agent population (older as well as younger speakers). As shown by e.g. Stanford and Kenny (2013), agent-based models can be sensitive to imbalances in the input data, such as unequal numbers of agents per agent group or disproportional amounts of input from single agents. However, further simulations provided in section S2 in the Supplementary Material show that the results on NZE presented in Section 4.3 were not a mere matter of imbalanced data.

A further (and unexpected) finding from the NZE simulations was the emergence of new phonological classes as a consequence of interaction. More specifically, as the interactions progressed, there was both

¹¹ Since 32 out of 48 word classes were associated with group 2 in Table 1, the chance level of agreement between these groups and sub-phonemes is $32/48 \approx 0.67$.

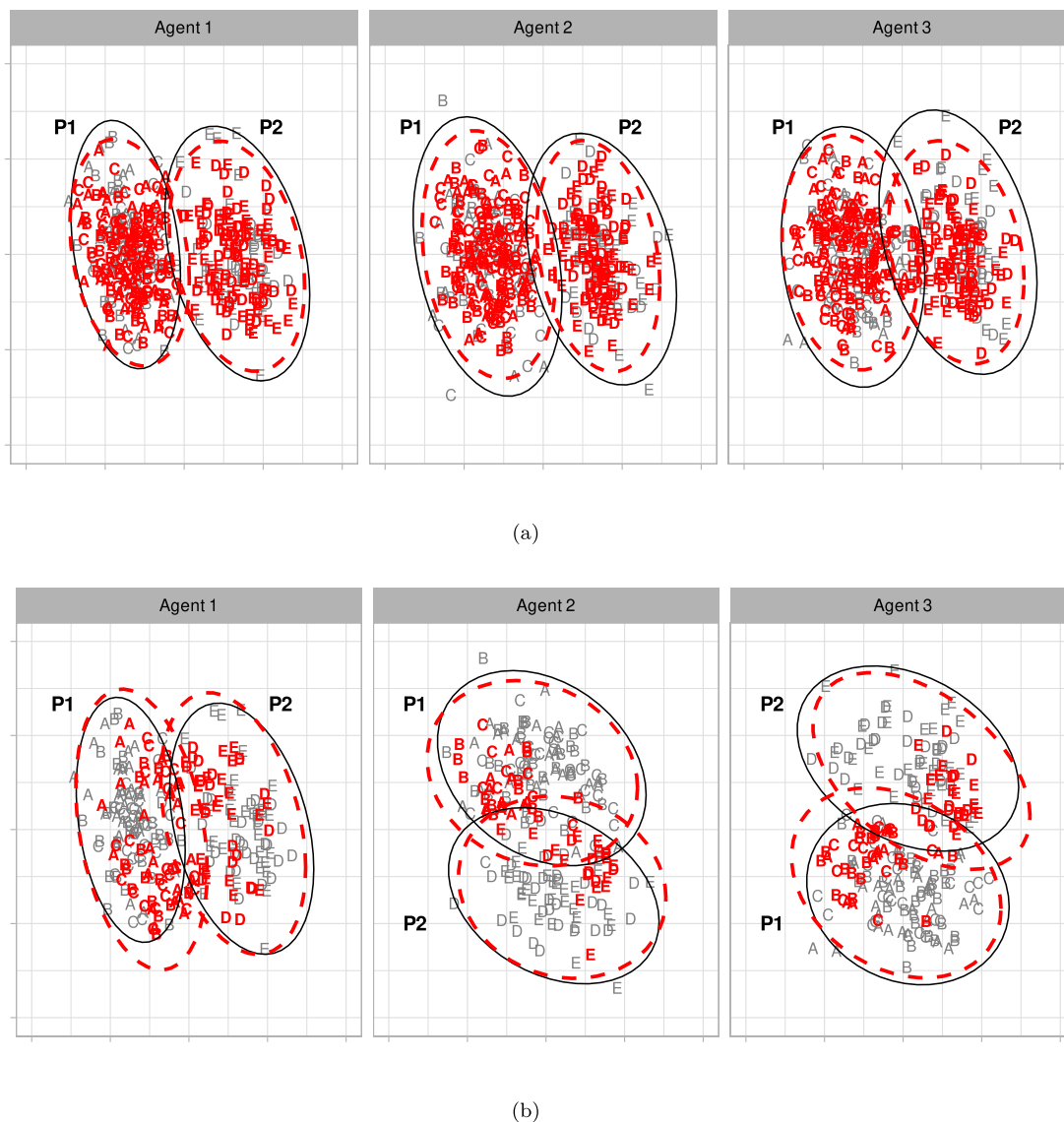


Fig. 8. A distribution of five words (A-E: 30 exemplars per word and per agent) in a two-dimensional space that were categorised as two phonological classes P1 and P2. Grey letters and black ellipses denote the initial conditions prior to the ABM simulation. Red letters and dashed ellipses are the conditions after interaction. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

an increase in the degree of the phonological /eə, ɪə/ merger and in the agreement across agents in their grouping of the centring diphthongs into two new phonological categories. These phonological categories were context-dependent, with one comprising mostly words in which the diphthong was followed by an alveolar (*beard, cares, merely*) and the other including primarily pre-pausal or pre-/r/ (*caring*) contexts. The statistical analysis showed that this context-dependent grouping of exemplars into sub-phonemes was already evident at the baseline, but became more apparent towards the end of the simulation. The likely reason why these new sub-phonemic classes emerged was because of a coarticulatory effect that was consistent across agents (Gubian et al., 2019). More specifically, when the centring diphthongs occur before an alveolar, the F2 frequency that is already high because of the first /ɪ, e/ component is likely to remain high due to the high F2-locus at just under 2 kHz of alveolars (Liberman et al., 1954; Kewley-Port, 1982; Fant, 1973). By contrast for pre-/r/ and pre-pausal diphthongs, F2 and especially F3 are likely to be lower. Because there is no rapid transition from a high vowel to a following consonantal constriction in pre-pausal position, the vowel is longer and has more time to attain a phonetically central position which are associated with a lower F2 and F3 than preceding alveolars. Because this is a consistent coarticulatory effect

across agents, two sub-phonemic classes emerged for the population that map to these two sets of words (also see Section 5). Additional simulations in section S2 in the Supplementary Material showed that the presence of the coarticulatory effect in the NZE data and its subsequent phonologisation was not causally related to the merger of /ɪə/ and /eə/.

5. General discussion

The aim of this paper was to present the IP-ABM and test it on two different sound changes. The first of these was /u/-fronting which is a phonetic change that does not affect the phonological system of the language; the second was a merger between two centring diphthongs requiring a flexible association between acoustic exemplars and phonological classes. As explained in Section 2.2, the phonology module implemented in the IP-ABM consisted of two fully unsupervised, general-purpose learning algorithms and was shown to be able to model both stability and change at the phonological level (Sóskuthy, 2015).

The application of the IP-ABM to /u/-fronting showed a similar pattern of results to that in Harrington and Schiel (2017), i.e., an

asymmetric change such that the retracted /u/ from the older speakers shifted more towards the front of the younger speakers' vowel spaces than vice versa. Although the derivation of phonological classes in the IP-ABM was entirely bottom-up and data-driven, the simulations with /u/-fronting showed no changes to phonological categorisation. This finding demonstrates, therefore, that a split, merge, or other type of phonological re-organisation is not an inevitable outcome following interaction in the IP-ABM.

When the ABM was applied to the New Zealand English data, interaction increased the degree of merger between the centring diphthongs. The merger consisted of the reduction of acoustic differences between /ɪə/ and /eə/ along with the neutralisation of the phonological distinction between the two. This increase in merger during the simulation was to a certain extent caused by a high degree of inconsistency across agents in the way that words were distributed in an acoustic space. An example of a consistent relationship is in the data from /u/-fronting. The consistency across agents in this case is that /i/-words (*heed, feed*) and /u/-words (*who, food*) were positioned acoustically in phonetically front vs. central or back parts of the vowel space for all agents. For /ɪə, eə/ by contrast, there was no such consistent relationship prior to interaction. Thus, although /ɪə, eə/ words differed to a certain extent in phonetic height for some agents (although always with some overlap between them), for other agents this was not necessarily the case. Consequently, the consistency with which words were acoustically positioned differed across agents.

The evidence that such an inconsistency contributes to the merger is further illustrated in the simulations in Fig. 8. In both 8(a) and 8(b), the letters A-E represent different words. Fig. 8(a) shows the outcome when the positioning of words in an acoustic space is consistent across agents. In this case, there are two phonological categories, P1 and P2, that map to words $P1 \in \{A, B, C\}$ and $P2 \in \{D, E\}$. Moreover, P1 is positioned to the left of P2 on the x-axis for all three agents. The outcome of applying the ABM to these artificially created data in Fig. 8(a) was the preservation of the phonological distinction between P1 and P2. The reason why the phonological separation was preserved was because the perceived, incoming acoustic exemplars that agents received from each other were usually positioned within the same phonological category: for example, agent 1 received exemplars of A, B, C from agents 2 and 3 that typically fell somewhere around the centre of agent 1's acoustic space in which A, B, C were located (i.e. near the centre of P1). Consequently, interaction typically preserved the separation between the two phonological categories, even though, as Fig. 8(a) shows, there was a certain degree of overlap between them.

The simulation in Fig. 8(b) shows the outcome when there is an inconsistent relationship across agents in how words are positioned in an acoustic space. In Fig. 8(b), there were the same two phonological categories consisting of the same sets of words and the degree of overlap between P1 and P2 was also much the same as in Fig. 8(a). But whereas in Fig. 8(b) P1 and P2 are distinguished by the x-axis parameter for agent 1, for agents 2 and 3 it is the y-axis parameter that separates them; in addition, P1 is above P2 on the y-axis for agent 2, but below P2 for agent 3. Interaction between the agents in Fig. 8(b) caused an increase in the overlap between the two phonological categories. This happened because perceived exemplars from other agents were typically positioned at the periphery of a phonological category. For example, agent 1 in Fig. 8(b) received many incoming A, B, C words at the top of the P1 ellipse from agent 2 and at the bottom of the P1 ellipse from agent 3. Because these perceived tokens fell within agent 1's P1 ellipse, they were memorised. Because they were at the ellipse periphery, the effect of memorisation was to broaden the ellipse. A broadening of the ellipses, which happens for the same reason to both P1 and P2 and for all three agents, is the pre-condition for a merger to be possible. Thus, to summarise: an inconsistent positioning of the same words in an acoustic space across agents (Fig. 8(b)) often leads to a merger because such inconsistency causes a broadening of phonological categories during interaction.

This type of inconsistency between speakers has been observed for some types of so-called marginal phonological relationships (Hall, 2013; Ladd, 2006; Nadeu and Renwick, 2016; Scobbie and Stuart-Smith, 2008), that have been characterised as intermediate between unambiguous phonological contrasts and neutralised forms. An example of an intermediate phonological relationship concerns the status of mid vowels in many Romance languages which have been neutralised within the front /e, ε/ and within the back /o, ɔ/ vowel sets (Recasens and Espinosa, 2009). In some varieties of Catalan by contrast, the distinction is maintained (Nadeu and Renwick, 2016). However, these vowel pairs differ from many other types of phonological contrast, not only because the functional load due to the distinction is small, but also because speakers even of the same dialect often disagree about how /e/ vs. /ε/ or /o/ vs. /ɔ/ divide up the vocabulary. A similar finding is reported in Renwick and Ladd (2016, p. 23) for Italian who show “significant lexical variability between speakers in the choice of mid vowel in stressed syllables” and that “speakers’ productions do not always match their judgment about which vowel they use in which word”. Just this type of inconsistency in how phonological categories map to lexical items across speakers is predicted by the simulation in Fig. 8(b) to be a prior stage along the path towards complete neutralisation that is found in many other varieties of Romance languages.

Finally, the agent-based model developed in this study and the associated model of sound change are founded upon the idea that the phonology that links acoustics and the lexicon is stochastic and speaker-specific. Because of feedback connections between different levels of the model (Hay et al., 2015; Pierrehumbert, 2001, 2002, 2003; Wedel, 2006, 2012), phonological categories are incrementally updated when new exemplars have been memorised. A speaker-specific phonology means that speakers even of the same spoken accent do not necessarily carve up the lexicon into identical phonological categories. For the reasons outlined above, this speaker-specific phonology is one of the factors that can contribute to sound change. This view that phonology is stochastic and can vary between speakers is predicted by episodic models of speech (Beckner et al., 2009; Goldinger, 1996; Pierrehumbert, 2003, 2006; Johnson, 2006). This is because phonology emerges as a stochastic generalisation over lexical items which are in turn formed from memorised speech signals: these collections of exemplars are necessarily idiosyncratic because their accumulation is governed by the stochastic nature of a person's interlocutors and frequency of interaction. This is also consistent with many emergentist views of language (Bybee, 2008, 2002; Little et al., 2017; Lindblom, 1998; Studdert-Kennedy, 1998) in which the phonology of a population of speakers is shaped and updated as a consequence of spoken interactions between individuals that form part of the population (e.g. Beckner et al., 2009; Kirby and Sonderegger, 2013).

CRedit authorship contribution statement

Michele Gubian: Methodology, Software, Validation, Formal analysis, Writing – original draft, Writing – review & editing, Visualization. **Johanna Cronenberg:** Software, Validation, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Jonathan Harrington:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Algorithms

This section provides background information on three state-of-the-art machine learning algorithms that are employed in the IP-ABM, namely GMMs, NMF, and SMOTE.

A.1. Gaussian Mixture Models-based clustering and classification

A Gaussian Mixture Model (GMM) is a parametric probability density function represented as a weighted sum of Gaussian component densities (Reynolds, 2009). Its general form is:

$$f(x|\lambda) = \sum_{j=1}^G w_j \cdot g(x|\mu_j, \Sigma_j), \quad (5)$$

where x is a continuous-valued d -dimensional vector, G is the number of components, $g(\cdot)$ is the multidimensional Gaussian density function and $\lambda = \{w_j, \mu_j, \Sigma_j\}$, $j = 1, \dots, G$ indicates the set of weights w_j satisfying $\sum_{j=1}^G w_j = 1$, the d -dimensional mean vectors μ_j and the $d \times d$ -dimensional covariance matrices Σ_j .

Given a set of N d -dimensional data vectors $\{x_i\}$, $i = 1, \dots, N$, and fixing the number of components G , a model like Eq. (5) can be obtained by maximum likelihood (ML) estimation, usually applying the expectation–maximisation (EM) algorithm, as there is no closed-form solution for the maximisation of Eq. (5) with respect to λ . Often constraints are applied on the structure of Σ_j , e.g. diagonal, or parameters are tied across components, e.g. $\Sigma_j = \Sigma, \forall j$. The number of components can be estimated by first estimating several candidate models like Eq. (5), each one with a different number of components G , and then applying the Bayesian information criterion (BIC), or alternative IC methods, to select the best model.

Eq. (5) can be used as a clustering model for the data vectors $\{x_i\}$ that were used to estimate its parameters. By interpreting the mixture components as clusters, any vector x , from the training dataset or otherwise, can be assigned to a cluster j by applying the maximum a posteriori (MAP) criterion:

$$Cluster(x) = \arg \max_j \frac{w_j \cdot g(x|\mu_j, \Sigma_j)}{\sum_{k=1}^G w_k \cdot g(x|\mu_k, \Sigma_k)} = \arg \max_j w_j \cdot g(x|\mu_j, \Sigma_j). \quad (6)$$

When information on class membership of the data is available, i.e. the vectors x_i are paired with corresponding class labels $y_i \in \{1, \dots, K\}$, where K is the number of classes, it is possible to use the form Eq. (5) as building block for a hierarchical classification model (Mixture Discriminant Analysis, MDA, Fraley and Raftery, 2002). MDA is a generalisation of Linear and Quadratic Discriminant Analysis (LDA, QDA) that allows the density of each class to be of the form of Eq. (5). Class membership y for a new vector x is obtained by applying Bayes's rule:

$$\begin{aligned} Class(x) &= \arg \max_y Pr(x \in y) = \arg \max_y \frac{\tau_y \cdot f_y(x|\lambda_y, G_y)}{\sum_{k=1}^K \tau_k \cdot f_k(x|\lambda_k, G_k)} \\ &= \arg \max_y \tau_y \cdot f_y(x|\lambda_y, G_y), \end{aligned} \quad (7)$$

where $\{\tau_y\}$, $y \in \{1, \dots, K\}$ are the proportions of members of class y in the training set, $f_y(\cdot)$ are functions of the form of Eq. (5), each having in general a different parameter set λ_y and number of components G_y .

In the IP-ABM code base, both GMM clustering and MDA classification are implemented using the R package *mclust* (Scrucca et al.,

2016). The number of acoustic components, as well as a suitable set of constraints on the parameter set λ , are determined by the BIC criterion as implemented by the function *Mclust*. The MDA model is estimated using function *MclustDA*, where the number of components of each phonemic class is fixed according to the mapping obtained from the NMF procedure (see Appendix B). For example, if clusters a_1 and a_2 map to sub-phoneme p_1 and a_3 maps to p_2 , then the number of components for p_1 and p_2 are fixed to 2 and 1, respectively. This helps to preserve the original purely acoustic-based structure expressed by the GMM clustering model.

A.2. Non-negative Matrix Factorisation

Non-negative Matrix Factorisation (NMF, Seung and Lee, 2001) is a family of algorithms that given a $m \times n$ input matrix C with non-negative entries determines an approximation of it in the form of a product of two matrices, W and H , also with non-negative entries. Formally:

$$C \approx W \cdot H, \quad (8)$$

where W has dimension $m \times r$, H has dimension $r \times n$, and usually $r < \min(m, n)$. Once r is fixed, Eq. (8) is solved by minimising the difference between C and its approximation $W \cdot H$, i.e. W and H are the solution to:

$$\min_{W, H} \|C - W \cdot H\|_F^2, \text{ such that } W, H \geq 0, \quad (9)$$

where $\|\cdot\|_F$ is the Frobenius norm, the analogous of Euclidean norm for matrices.

In applications, each of the n columns c_j of C represents an observation or feature vector, where the m elements of c_j represent counts, frequencies, energies or other non-negative quantities. For example, in text mining the n columns of C represent documents and the m rows word counts or frequencies (Shahnaz et al., 2006). As the number of words $\times$ documents can be in the thousands $\times$ millions, it is convenient to find a compact approximation of C in the form of Eq. (8) with r as small as possible, say in the tens. Similarly to PCA, the columns of W can be seen as bases or principal components capturing the fundamental traits of the columns of C (though the columns of W are not constrained to be orthogonal), while each column h_j of H determines the linear combination of columns of W that best reconstructs a corresponding input column c_j . Differently from PCA, the non-negativity of W and H in Eq. (8) allows to interpret the composition of columns of W as strictly additive contribution from each basis in the quantities determined by the positive coefficients in H , where no contribution can undo (i.e. subtract) parts of the others. For example, in text mining the r columns of W are interpreted as topics, and each document c_j , codified by the number of occurrences of m selected words (i.e. bag of words representation), is a weighted sum of topics, the weights being the elements of column h_j of H .

A further step in the interpretation of Eq. (8) is achieved whenever W and H are sufficiently sparse, in which case the columns of H tend to ideally have only one large entry, which identifies only one column of W . This means that each of the n input column vectors of C can be approximated by just a coefficient multiplying one of the $r < n$ columns w_k of W , i.e. $c_j \approx H(k, j) \cdot w_k$ when $H(k, j) \gg H(i, j), \forall i \neq k$. The particular column w_k approximating a given input vector c_j can be interpreted as its centroid or prototype, analogously to other prototype-based clustering algorithms, e.g. k-means (see Ding et al., 2005, for a formal statement). This interpretation allows to utilise NMF as a clustering algorithm, with r being the number of clusters. In order to improve the performance of NMF as clustering method, several sparse solutions have been proposed (e.g. Kim and Park, 2008; Pascual-Montano et al., 2006). As for most clustering algorithms, the optimal number of clusters r has to be determined by applying a model selection criterion external to the algorithm itself.

In the IP-ABM code base, NMF clustering is implemented using the R package *NMF* (Gaujoux and Seoighe, 2010). In particular, the NMF

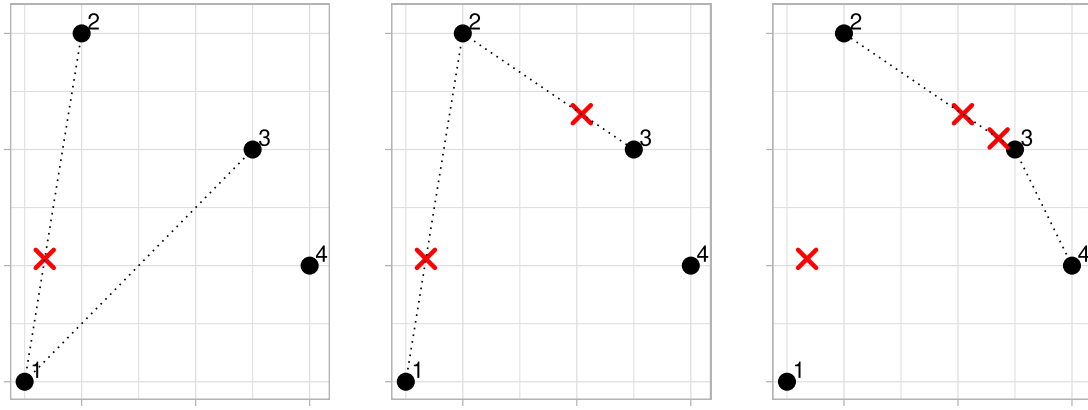


Fig. 9. Example of application of SMOTE to a set of four points (numbered). Each panel shows the generation of one new point (red 'x's), in succession from left to right. In each panel, a pivot point (point 1, 2, and 3 from left to right) is connected by segments to its $k = 2$ nearest neighbours. A new point is generated at a random position on one of those segments. The new points are not used in the generation process, i.e. they do not become pivot points nor are they considered as neighbours. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

algorithm version by Pascual-Montano et al. (2006) is selected, which produces sparser solutions (setting method option to 'nsNMF' in the nmf command). The cluster assignment is determined by taking the arg max of each column of matrix H .

A.3. SMOTE

Synthetic Minority Over-sampling TEchnique (SMOTE, Chawla et al., 2002) is a non-parametric resampling algorithm used to mitigate the negative impact of imbalanced datasets on supervised classification. SMOTE has been applied to several classification problems (see Fernandez et al., 2018, for a survey), its performance and theoretical properties have been evaluated (Blagus and Lusa, 2013), and a number of modifications to the original algorithm have been proposed (Fernandez et al., 2018; Leevy et al., 2018). In its basic form, SMOTE generates extra artificial data for the under-represented class(es) in such a way as to preserve their original statistical properties. For example, if class A contains only 10 data points, while class B has 100, SMOTE can be applied to class A to top it up to the level of class B by generating 90 artificial data points. The generative process only makes use of the existing data points from the under-represented class. In other words, SMOTE does not require the existence of a numerous class to generate extra data for a less numerous class. As a consequence, SMOTE can be applied more generally to increase the number of data points, e.g. when data are insufficient (though possibly balanced) for a given task.

The data generation mechanism of SMOTE is illustrated in Fig. 9. Suppose there are n data points living in a d -dimensional space ($n = 4$ numbered points in a $d = 2$ -dimensional space in Fig. 9) and n_S new points need to be created. Each new data point (red 'x's in Fig. 9) is generated by selecting a pivot point from the original n data points, connecting it to its k nearest neighbours ($k = 2$ in Fig. 9), selecting one of such connecting segments at random and generating a new data point along that segment at a random location. The process is repeated n_S times, each time changing the pivot point. Fig. 9 shows the generation of the first three points. Both k and n_S are parameters to be set by the user.

Appendix B. Design and implementation of the IP-ABM

B.1. Production-perception loop

The implementation of token production as well as the representation of sub-phonemic classes involve the use of density distributions. We opted for Gaussians and Gaussian mixture models because of their

generality. In particular, for production we opted for a unimodal Gaussian distribution, rather than a general GMM, because usually the number of tokens available for the estimation is quite reduced. Sub-phonemic classes are represented as a Mixture Discriminant Analysis model (MDA, Fraley and Raftery, 2002), a hierarchical model where each class is a GMM. For example, the listener panel in Fig. 3 represents an MDA model with three classes p_1 – p_3 , where p_1 is a two-component GMM, p_2 and p_3 are one-component GMMs, i.e. simple Gaussians (see Table 2 for a list of symbols used in this section). More details on GMMs and MDA are provided in Appendix A.1. Typicality and discriminability tests involve some form of distance between a token and sub-phonemic classes. Typicality is operationalised by the Mahalanobis distances (Mahalanobis, 1936) between the token and all the GMM components belonging to the relevant sub-phoneme. A threshold for the test was empirically set to a value corresponding to accepting tokens that lie within the mean-centred ellipsoid containing 95% of the Gaussian component probability mass. When the sub-phoneme consists of more than one component, the distances to all components are computed and the threshold is applied to the smallest distance. For example, the token received by the listener in Fig. 3 passes the typicality test because it lies inside the 95% probability mass ellipsoid of its class p_2 , while it would fail if it belonged to class p_1 or p_3 . The discriminability test is implemented by computing the maximum a posteriori probability (MAP) of the received token according to the MDA model. The test is passed if the sub-phonemic class the token belongs to according to its word label coincides with the MAP class; intuitively, the test is passed if the token is not acoustically ambiguous. For example, in Fig. 3 the received token (x, w_1) , whose word label w_1 belongs to p_2 , will pass the discriminability test if the posterior probability of x belonging to p_2 is higher than for the other two classes. More formally, a new exemplar x , which according to phonology, i.e. via word membership $p(w(e_i))$, belongs to phoneme y , is memorised only if Eq. (7) also yields y as class membership. Note that the MDA model is based on class memberships obtained via the chain $p(a(e_i))$, which in general does not yield the same result as the one that uses word membership, i.e. $p(w(e_i))$. The reason is that acoustic clusters contain impurities, i.e. exemplars that do not belong to the sub-phonemic class that is present as majority in a given acoustic cluster. We prefer to use this 'impure' association to build the classification model because this represents the acoustic counterpart of the sub-phonemic organisation of words, hence should be based on acoustic similarity rather than on pure word membership.

B.2. Derivation of acoustic and sub-phonemic representations

B.2.1. Overview

Each agent periodically re-estimates its acoustic and sub-phonemic models following a two-stage process schematised in Fig. 10. First, a

Table 2
Symbols used in Appendix B.

Symbols	Meaning
x	An acoustic vector, e.g. $[1.2, -0.4, 0.88]^T$, when the acoustic space is 3-dimensional
d	Number of dimensions of the acoustic space
w	A word type and a specific position within the word, e.g. <u>food</u>
e	An exemplar, i.e. a tuple (x, w)
$x(e)$	Acoustic vector of exemplar e
$w(e)$	Word type of exemplar e
N_e	Number of exemplars in an agent's memory
N_a	Number of acoustic clusters
N_p	Number of sub-phonemic classes
N_w	Number of word types
a	An acoustic cluster, determined by probabilistic membership to a Gaussian mixture component
p	A sub-phonemic class, determined by applying NMF to acoustic clusters
C_a	$N_w \times N_a$ matrix of exemplar counts, $C_a(j, k)$ is the count of exemplars of word w_j that belong to acoustic cluster a_k
C_p	$N_w \times N_p$ matrix of exemplar counts, $C_p(j, k)$ is the count of exemplars of word w_j that belong to sub-phonemic class p_k ; these counts allow impurities. The sub-phonemic class of word type w_j is determined by picking $\text{argmax}_k C_p(j, k)$
$a(x(e)) = a(e)$	Acoustic cluster associated to an acoustic vector
$p(a(e)) = p(a)$	Sub-phonemic class associated to an acoustic cluster
$p(w(e)) = p(w)$	Sub-phonemic class associated to a (position within a) word
Relations	Comment
$1 \leq N_p \leq N_a$	Highest number of sub-phonemic classes occurs when each acoustic cluster constitutes a phonological class
$p(a(e)) \neq p(w(e))$	If an exemplar is 'impure', the sub-phonemic class associated to its location in the acoustic space is different from that associated to the word it belongs to

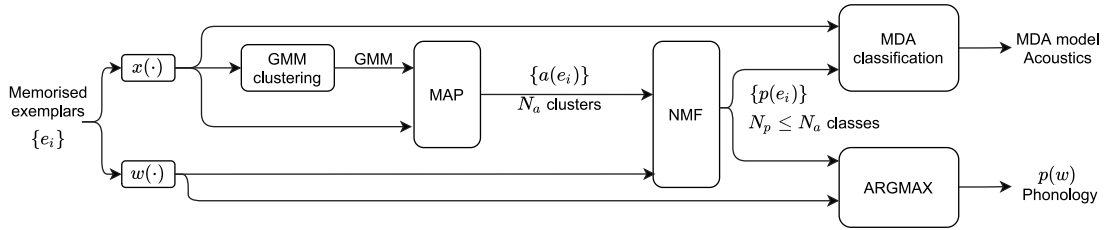


Fig. 10. General scheme of the implementation of acoustic and sub-phonemic representations in the IP-ABM.

GMM clustering model as in Eq. (5) is estimated from the acoustic representation of all the $N = N_e$ exemplars stored in memory, i.e. the set $\{x(e_i)\}$, $i \in \{1, \dots, N_e\}$, disregarding any information on sub-phonemic classes established in previous estimations. The number of acoustic clusters $N_a = G$ is estimated by BIC. By applying Eq. (6) each exemplar is assigned to an acoustic cluster $a \in \{1, \dots, N_a\}$ corresponding to a single Gaussian component. The acoustic cluster membership information associated to each exemplar $\{a(e_i)\}$, but *not* the corresponding acoustic representation $\{x(e_i)\}$, is fed to the NMF-based procedure, which combines it with the word membership $\{w(e_i)\}$ to determine which acoustic clusters shall be joined together, i.e. a mapping from the acoustic labels $\{1, \dots, N_a\}$ to the sub-phonemic labels $\{1, \dots, N_p\}$, $N_p \leq N_a$. This mapping allows to indirectly associate each exemplar with a sub-phonemic class, i.e. $p(a(e_i))$. Using these class assignments, a MDA classification model is estimated, where the number of classes coincides with the number of sub-phonemic classes estimated by NMF, i.e. $K = N_p$ in Eq. (7). NMF operates on a matrix where each cell contains the exemplar count of a given word (row) in a given acoustic cluster (column). The result is a grouping of columns having similar count patterns, which are then merged together. Sub-phonemic classes are represented by those columns, while each word is assigned to a class by picking the column with the highest count for that word.

The two-stage process described above is executed periodically and independently by each agent. Within the same agent, each execution

bears no memory of the acoustic and phonological representations derived in previous executions, a choice driven by implementation convenience. Continuity is indirectly guaranteed by making sure that only a small fraction of memory has changed (through exemplar memorisation and forgetting) between one execution and the next.

B.2.2. How NMF identifies sub-phonemic classes

The input to NMF are (i) the set of acoustic cluster memberships $\{a(e_i)\}$, $i \in \{1, \dots, N_e\}$, obtained by applying MAP as in Eq. (6) to the acoustic exemplars $\{x(e_i)\}$, and (ii) the set of word memberships of each exemplar, $\{w(e_i)\}$. These two pieces of information are combined into a $N_w \times N_a$ matrix of counts C_a , whose (j, k) element is the count of exemplars of word w_j that belong to acoustic cluster a_k , i.e. $C_a(j, k) = |\{i : w(e_i) = w_j, a(e_i) = a_k\}|$. This matrix is decomposed into the product of two matrices using NMF as in Eq. (8), i.e. $C_a \approx W \cdot H$, where the rank of the approximation corresponds to the number of sub-phonemic classes ($r = N_p$) and it is empirically determined as explained later on. Matrix H is further processed to obtain an indicator matrix $\tilde{H}$, where each column h_k of H is substituted with a column of zeros and ones, with a single 1 at the position corresponding to the maximum element of h_k and zeros elsewhere. This is then used to recompute word counts according to the new N_p sub-phonemic clusters:

$$C_p = C_a \cdot \tilde{H}^T, \quad (10)$$

where C_p is a $N_w \times N_p$ count matrix and T is matrix transposition.

An example taken from the SSBE dataset (Section 3) is illustrated as follows. One of the 22 agents at simulation start has the acoustic cluster counts C_a shown in Table 3, where we see that this agent divides its acoustic space into $N_a = 5$ clusters, a_1 to a_5 , corresponding to the five Gaussian mixture components plotted in Fig. 11(a). Applying Eq. (8) to C_a , where we fix the rank to $r = N_p = 3$, we obtain:

$$\underbrace{\begin{bmatrix} 0 & 0 & 2 & 7 & 0 \\ 0 & 3 & 0 & 0 & 7 \\ 9 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 10 & 0 \\ 0 & 9 & 0 & 0 & 1 \\ 8 & 0 & 2 & 0 & 0 \\ 0 & 7 & 0 & 0 & 3 \\ 9 & 0 & 1 & 0 & 0 \\ 0 & 9 & 0 & 0 & 1 \\ 3 & 0 & 6 & 1 & 0 \\ 0 & 0 & 0 & 10 & 0 \end{bmatrix}}_{C_a} \approx \underbrace{\begin{bmatrix} 0 & 0 & 0.31 \\ 0 & 0.25 & 0 \\ 0.25 & 0 & 0 \\ 0 & 0 & 0.34 \\ 0 & 0.25 & 0 \\ 0.25 & 0 & 0 \\ 0 & 0.25 & 0 \\ 0.25 & 0 & 0 \\ 0 & 0.25 & 0 \\ 0.25 & 0 & 0 \\ 0 & 0 & 0.34 \end{bmatrix}}_W \times \underbrace{\begin{bmatrix} 29 & 0 & 12 & 0 & 0 \\ 0 & 28 & 0 & 0 & 12 \\ 0 & 0 & 0 & 28 & 0 \end{bmatrix}}_H, \quad (11)$$

where we note that the NMF solution is indeed sparse. For example, the first column of C_a is approximated by the first column of W multiplied by 29, the second column of C_a by the second column of W multiplied by 28, etc., each column of C_a is approximated by just one column of W .¹² This provides the foundation for using the NMF expansion as a clustering criterion, namely columns of C_a approximated by the same column of W , such as columns 1 and 3, belong together because they are approximately proportional, which in our application means that they roughly contain the same word types. By observing C_a in Table 3 we can see that this is the case for acoustic clusters a_1 and a_3 , which makes them good candidates for a merge. The merge of acoustic categories is obtained by constructing the indicator matrix $\tilde{H}$ from H :

$$\tilde{H} = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix},$$

the rows of which indicate with 1s the columns of C_a which should be merged (summed) into columns of C_p , e.g. the first row indicates that the first and third column of C_a will be summed together to become the first column of C_p . This operation is formalised by Eq. (10), the result of which is Table 3.

The result in Eq. (10) defines the mapping from acoustic clusters to sub-phonemic clusters $p(a(e_i))$, which is not yet a definition based on word types, rather on word exemplars. For example, according to Table 3 two exemplars of *cooed* belong to sub-phonemic class p_1 and seven to p_3 . To obtain a many-to-one mapping from word types (and position within word) to sub-phonemic classes we assign each word w to the sub-phonemic cluster with highest exemplar count, i.e.

$$p(w) = \arg \max_p C_p(w, p). \quad (12)$$

For example, $p(\text{cooed}) = p_3$, $p(\text{feed}) = p_2$, etc., as indicated by boldface figures in Table 3, by labels in Table 3 and by colours in Fig. 11(b). We call the entities defined by the mapping in Eq. (12) sub-phonemic *categories* or *classes*, to distinguish them from the sub-phonemic *clusters* identified by exemplar counts in C_p . This distinction is crucial as it brings along the concept of cluster *purity*, which plays a central role in determining the optimal number of sub-phonemic clusters, as explained below.

Purity is defined for each cluster as the fraction of exemplars that belong to that cluster, according to the majority rule defined in

Table 3

Exemplar counts by word type (rows) and by acoustic cluster (columns, 3a) or by phonemic cluster (columns, 3b). This is a snapshot of the memory of one of the agents from the /u/-fronting case reported in Section 3 taken at simulation start. In 3b boldface figures indicate which sub-phonemic cluster contains the majority of exemplars for each word type, which determines the assignment of words to sub-phonemic classes $p(w)$ in 3c.

(a) C_a					
Word	Acoustic clusters				
	a_1	a_2	a_3	a_4	a_5
<i>cooed</i>	0	0	2	7	0
<i>feed</i>	0	3	0	0	7
<i>feed</i>	9	0	1	0	0
<i>food</i>	0	0	0	10	0
<i>heed</i>	0	9	0	0	1
<i>hewed</i>	8	0	2	0	0
<i>keyed</i>	0	7	0	0	3
<i>queued</i>	9	0	1	0	0
<i>seep</i>	0	9	0	0	1
<i>soup</i>	3	0	6	1	0
<i>who'd</i>	0	0	0	10	0

(b) C_p			
Word	Sub-phonemic clusters		
	p_1	p_2	p_3
<i>cooed</i>	2	0	7
<i>feed</i>	0	10	0
<i>feed</i>	10	0	0
<i>food</i>	0	0	10
<i>heed</i>	0	10	0
<i>hewed</i>	10	0	0
<i>keyed</i>	0	10	0
<i>queued</i>	10	0	0
<i>seep</i>	0	10	0
<i>soup</i>	9	0	1
<i>who'd</i>	0	0	10

(c) $p(w)$	
Word	Sub-phonemic classes
<i>cooed</i>	p_3
<i>feed</i>	p_2
<i>feed</i>	p_1
<i>food</i>	p_3
<i>heed</i>	p_2
<i>hewed</i>	p_1
<i>keyed</i>	p_2
<i>queued</i>	p_1
<i>seep</i>	p_2
<i>soup</i>	p_1
<i>who'd</i>	p_3

Eq. (12), divided by its exemplar count. Purity of the whole clustering solution is defined as the minimum purity value among the clusters (other definitions are possible, e.g. taking the average instead of the minimum, cf. Manning et al. (2008)). Formally:

$$Purity = \min_k \left(\frac{|\{i : p(a(e_i)) = p(w(e_i)) = p_k\}|}{|\{i : p(a(e_i)) = p_k\}|} \right) \quad (13)$$

To illustrate, the purity of each cluster in Table 3 is the sum of boldface counts divided by all counts in each column, e.g. $Purity(p_1) = \frac{39}{41} = 0.951$, which is the smallest value among the three ($Purity(p_2) = 1.0$ and $Purity(p_3) = \frac{27}{28} = 0.964$), so also the purity of the whole clustering solution according to Eq. (13). The optimal number of sub-phonemic clusters N_p is determined empirically by running several instances of NMF for each value of N_p from 2 to N_a , and choose the result that on average reaches the best trade-off between a high number of clusters and a high purity. This criterion in general disfavors the

¹² The reader unfamiliar with matrix multiplication may consult any text on linear algebra.

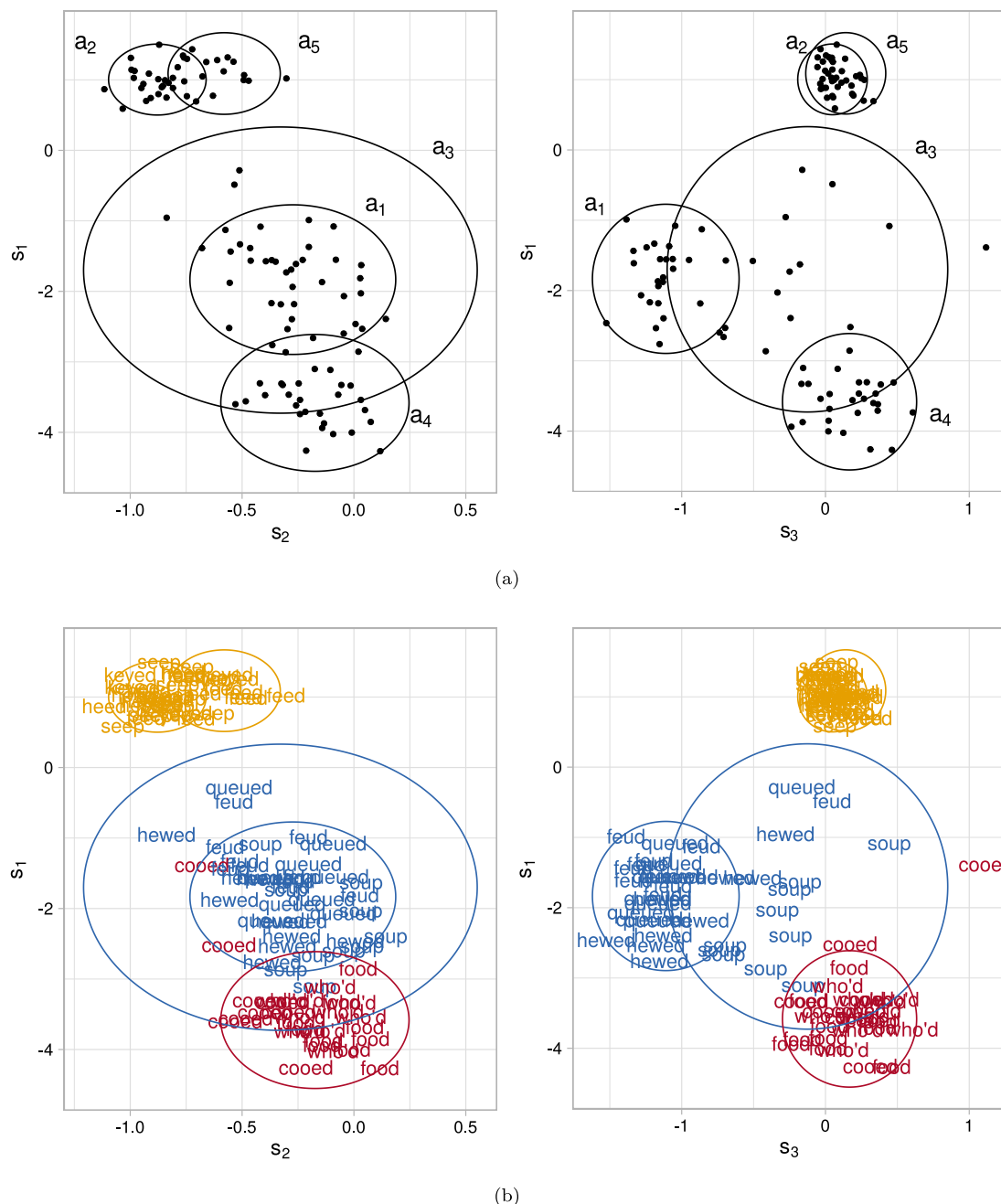


Fig. 11. Example of acoustic clusters 11(a) and sub-phonemic classes 11(b) in an agent. In 11(a), dots represent memorised exemplars in the acoustic space, and ellipses $a_1 - a_5$ identify acoustic clusters. In 11(b), the same exemplars are labelled according to the word they are associated with, while colours show their sub-phonemic classes. The acoustic space is three-dimensional (s_1, s_2, s_3) based on FPCA scores (Section 3.2.2); left and right views are projections on the (s_1, s_2) and (s_1, s_3) plane, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

two extreme solutions, i.e. $N_p = 1$ and $N_p = N_a$. On the one hand, the highest number of clusters $N_p = N_a$ is probably not the solution with the highest purity; on the other hand, the trivial solution $N_p = 1$, which always yields a purity of 1, has also the lowest number of clusters. The trade-off is regulated by a purity threshold θ_{purity} , which has to be set by the user. The optimal N_p is chosen to be the largest among those corresponding to an average purity above θ_{purity} . Note that the solution $N_p = 1$ can be chosen, provided that purity at $N_p = 2$ is below θ_{purity} . To illustrate, Table 4 shows how purity was computed for the example in Table 3 with Eq. (13) five times for each candidate number of sub-phonemic clusters N_p , from 2 to $N_a = 5$, and then the optimal

value $N_p = 3$ was chosen as the highest among the solutions above $\theta_{\text{purity}} = 0.75$.

B.3. Memory management

B.3.1. Overview

Two measures were implemented so that the agents’ memories are filled with an appropriate number of exemplars per word in the beginning and are kept at the same size throughout the simulation. In the case in which there are very few tokens from real speakers available for the initialisation of agents, the number of tokens can be increased by applying SMOTE (Chawla et al., 2002). This is a standard resampling

Table 4

Five computations of purity and their average value (columns) for different number of sub-phonemic clusters (rows). The optimal solution is $N_p = 3$, as its average is the one with the highest N_p among the solutions with average purity above $\theta_{\text{purity}} = 0.75$. The stochasticity of NMF allows for different solutions starting from the same input (rows). Missing values can occur whenever one or more clusters are empty, and such solutions are never chosen, as we consider them unstable.

N_p	Purity estimates					Average
2	0.95	0.95	1.0	1.0	0.9	0.96
3	0.95	0.95	0.95	0.95	0.95	0.95
4	0.58	0.50	0.50	0.50	0.50	0.52
5	0.50	–	0.50	0.50	0.50	–

technique that was adapted to the ABM architecture as explained in [Appendix A.3](#). When the agents start interacting and accepting each others' exemplars, their memories will grow in size. In order to control the memory size, the model allows agent listeners to forget exemplars, i.e. when an agent listener has accepted and memorised a new exemplar, another exemplar of the same word class is removed from their memory. The only constraint on forgetting is that agents cannot forget word classes, i.e., removal is blocked if the deletion of an exemplar would lead to a decrease in the number of exemplars of the word class with respect to the number of exemplars with which the word class was initialised.

B.3.2. SMOTE-based resampling

SMOTE has been employed in the IP-ABM as a robustness measure for production. As described in [Section 2.3](#), the generation of a new acoustic token for speech production is implemented by first selecting a word class, then using all the exemplars of that word class available in the agent's memory to estimate a Gaussian distribution. The new token is produced by extracting one sample from that distribution. When the number of exemplars (data points) $N_e(w)$ of the target word class w is insufficient for a reliable estimation of a (multi-dimensional) Gaussian distribution, i.e. $N_e(w) < N_{\text{prod}}$, where N_{prod} is a user-defined threshold, $N_{\text{prod}} - N_e(w)$ extra exemplars for the target class are generated on the fly via SMOTE. After estimation, the extra exemplars are discarded. In case $N_e(w) < k + 1$, i.e. if the number of available exemplars $N_e(w)$ is not even sufficient to identify k nearest neighbours, $k - N_e(w) + 1$ exemplars in the acoustic proximity of the $N_e(w)$ target exemplars from other word classes, but belonging to the same sub-phonemic class $p(w)$ as the target word, are added to the set of original data points used by SMOTE to generate new points.

The pure application of the above strategy was found to be insufficient as a protection against instability, especially at the beginning of ABM simulations. This is the case when very few tokens from real speakers are available for the initialisation of agents, so much so that not only production but also perception becomes unreliable, as the estimation of Gaussian mixtures necessary for the identification of acoustic clusters is also negatively affected by data scarcity. To mitigate the problem, a strategy to populate agent memories based on production, thus ultimately based on SMOTE, was introduced. A threshold $N_{\text{ex/word}}$ defining the minimum number of exemplars per word per agent available at all times throughout a simulation was introduced. For all cases not meeting such requirement at the beginning of a simulation, i.e. if $N_e(w) < N_{\text{ex/word}}$ for some word w , $N_{\text{ex/word}} - N_e(w)$ extra exemplars are generated by the same mechanism used for production and stored in the agent's memory. To exemplify, suppose $N_e(w) = 3$ and $N_{\text{ex/word}} = 10$ for a given agent and word w at simulation start. Then 7 extra exemplars are produced, each one applying the ordinary production algorithm based on the 3 original exemplars. If $N_e(w) < N_{\text{prod}}$, then SMOTE is applied each time, as explained above. Note that once a new exemplar is generated, it is not immediately made available as basis for the generation of the next ones. This is to avoid propagation of estimation errors. Only after all $N_{\text{ex/word}} - N_e(w)$ extra exemplars are generated, these are stored in memory and treated as

original exemplars once the actual simulation (i.e. agent interactions) begins. To keep $N_e(w) \geq N_{\text{ex/word}}$ throughout the simulation, a constraint on the forgetting process was imposed such that no exemplar from a word class w can be forgotten (removed) if as a result the number of exemplars for w would go below the threshold $N_{\text{ex/word}}$ (cf. [Section 2.3](#)). Note that $N_{\text{ex/word}}$ and N_{prod} are thresholds controlling different aspects, as the former is there to make sure that there is a minimum number of actual exemplars per word class in memory, which influences all aspects of the simulation, while the latter controls for the estimation basis for production only. In general $N_{\text{prod}} \geq N_{\text{ex/word}}$, for example if $N_{\text{ex/word}} = 10$ and $N_{\text{prod}} = 20$ it means that (i) $N_e(w) \geq 10$ for all w in all agents and (ii) SMOTE is applied in production whenever $10 \leq N_e(w) < 20$.

The SMOTE-related settings in the simulations presented in this work were fixed empirically to $N_{\text{prod}} = k + 1 = N_{\text{ex/word}} = 10$. This means in practice that no extra exemplars were generated at the beginning of the SSBE simulations ([Section 3](#)), as they already met the minimum exemplar requirement, while for the NZE simulations ([Section 4](#)) an aggressive initial re-population of agent memories was carried out, because the number of exemplars from real speakers per word per agent was usually only 1 or 2. In both cases, SMOTE was never applied during the simulations since $N_{\text{prod}} = N_{\text{ex/word}}$.

B.4. Implicit assumptions, proxies and simplifications

The IP-ABM makes use of a number of implicit assumptions and simplifications, both in the way each agent is modelled as well as in the way agents interact. Regarding the perception–production loop, first there is no explicit model of speech articulation. Any (co)articulation effect is indirectly modelled by the estimation of word-specific distributions, while sample extraction is a proxy for residual articulatory variation. Second, the transmission and perception of the acoustic properties of speech tokens between agents is assumed to be perfect. Besides not including any environmental acoustic noise, the IP-ABM bypasses the acoustic domain entirely by implementing the acoustic perception of a token as a mere copy of its acoustic parameters, which implies that the same acoustic parameterisation is applied by all agents. The concept of token contains in itself the further assumption that all agents are parsing the continuous stream of speech in the same way insofar as the isolation of speech sounds is concerned. Finally, word recognition is assumed perfect, an assumption reflected by the error-free copy of the lexical information between interacting agents.

Agents' mental representations are also shaped by simplifying assumptions. The lexicon, which is identical for all agents, is actually a flat list of labels, as no orthographic or morphological information is used by the agents. More specifically, a “word” here is intended as the identification of a speech sound belonging to a certain word in a certain position, e.g. the vowel in *food*. There is no representation of a word as a sequence of speech sounds, which implies that there is no explicit or emerging pressure to maintain minimal pairs (see also the discussion in [Section 2.2](#)). The lexicon merely reflects the list of words containing the sound(s) under investigation of which agents have acoustic exemplars stored in memory, which in practice limits the lexicon's size quite significantly. As a result, the emergence of sub-phonemic classes may be biased by the reduced variety of represented acoustic contexts. In the simulations presented here the lexicon is also assumed flat in terms of word frequency, although this is not a limitation of the IP-ABM in general, as it is straightforward to implement frequency-biased as opposed to uniform speech production.

There is no explicit notion of time. Implicitly, time advances at every interaction between two agents exchanging a token, which induces a partial ordering over interactions. In the current implementation, interactions occur only in pairs and only the agent listener can modify its state after an interaction by storing and/or forgetting an exemplar. Hence, if interaction 1 is a token exchange from agent A to B, interaction 2 from C to D, interaction 3 from C to E, and interaction

4 from B to F, then the only order constraint is that interaction 1 occurred before 4, because B as listener in 1 may have modified its state before acting as speaker in 4, while all other interactions may have occurred in any order, e.g. (1,2,3,4) is indistinguishable from (3,1,4,2). The notion of time based on interactions is not calibrated. Each interaction involves the exchange of a single token, which may or may not be memorised by the listener. The impact of one new exemplar in memory on the future production and perception depends to a first approximation on the amount of exemplars already present in memory, i.e. the more the exemplars, the smaller the relative impact of a single new exemplar will be, as both perception and production are operationalised by some form of probability distribution estimation on the exemplars. However, the number of exemplars in an agent's memory is primarily determined by the available speech data, i.e. neither determined on the basis of any estimation on the amount of speech episodes humans may retain in memory, nor calibrated in such a way as to reproduce an experimentally quantified effect on speech production, e.g. based on imitation experiments. As a consequence, there is no way to determine an equivalence between, say, 1000 interactions and an amount of time in a real community of speakers. On top of this, interactions between agents are reduced to the exchange of a single token, while human conversations obviously involve exchange of a comparably larger amount of speech material between two (or more) speakers.

Agents' memories are initialised with speech material from actual speakers, typically one speaker initialising one agent, although the same speaker's material may be used to initialise more agents using resampling techniques. As a consequence, the amount of speakers available in the data set, which can be as small as 10 or 20, determines the size of the modelled agent population. A small population can introduce artefacts in that the relative importance of single individuals is disproportionately large. This may be a problem when the agent population models a large community of speakers, like in the cases presented here, while it may be adequate when modelling an actual isolated community (Harrington et al., 2019). Groups of agents are obtained in two, possibly combined ways. First, groups of speakers of different accents or speaking styles are implicitly created by initialising agents with speech material from speakers of those distinct accents (Section 2.4), often using an apparent time paradigm, whereby older and younger speakers of the same language are used as proxies for two different stages of sound change, the younger representing the more advanced one. Second, groups within a population can be predefined explicitly and thus be considered by the rules imposed on interactions, e.g. by allowing only inter- or only intra-group interactions. These groups may or may not coincide with the accent-based groups implicitly defined at initialisation.

Appendix C. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.specom.2023.01.004>.

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